SMART BUS SYSTEM FOR REAL-TIME PASSENGER MANAGEMENT

(IoT-Based Real-Time Bus Tracking & ETA Prediction)

25-26J-267

Project Proposal Report

B.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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Project Proposal Report

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August 2025

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Public bus transportation serves as a vital mobility solution for millions of daily travelers in Sri Lanka, yet the sector remains challenged by late arrivals, overcrowding, limited accessibility for differently abled passengers, and inefficient fleet management. This research presents a modular Smart Bus System designed to address these challenges through the integration of advanced IoT, data analytics, computer vision, and inclusive mobile technologies. The proposed system comprises four core components: IoT-Based Real-Time Bus Tracking & ETA Prediction utilizing hybrid GSM and Wi-Fi connectivity with LSTM models for highly accurate arrival estimates; Vision-Based Passenger Counting and Occupancy Monitoring leveraging embedded deep learning and a hybrid sensor approach for anonymous, real-time crowd estimation; MobileBased Passenger Interaction and Accessibility System offering multi-modal user interfaces and demand-responsive routing to enhance inclusivity and communication; and a Central Fleet Management and Route Optimization Engine that employs dynamic, demand-aware route adjustments, predictive AI scheduling, and cross-modal trip planning with trains and ride-sharing services. Methodologically, our approach integrates real-time and historical transport data from sensors, GPS, user interactions, and traffic feeds into a unified analytical engine, enabling adaptive optimization and continuous schedule refinement. Anticipated outcomes include improved punctuality, optimized seat occupancy, enhanced passenger experience, and smarter resource use across multimodal transit networks. The study concludes with recommendations for scalable deployment in developing regions and outlines the expected impact on urban mobility, commuter satisfaction, and system-wide efficiency.

Keywords: Public Transport, IoT, Fleet Management, Data Analytics, Route Optimization, Inclusive Design, Urban Mobility

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Abbreviation

IT	Information Technology
GSM	Global System For Mobile Communication
Wi-Fi	Wireless Fidelity
LSTM	Long Short-Term Memory
AI	Artificial Intelligence
GPS	Global Positioning System
TC	Test Cases

1.0 INTRODUCTION

1.1 Background and Literature

For millions of metropolitan commuters, public transport systems are crucial. However, a major obstacle to dependability and customer pleasure is the unpredictable nature of arrival timings. In order to address this issue, GPS-based tracking is becoming more and more popular; nevertheless, the majority of current systems only use static algorithms or crude average speed estimates, which are unable to adjust to changing traffic, weather, or event conditions.

GPS, accelerometers, and other sensors can be used to collect high-frequency, real-time position and operating data from buses thanks to IoT technology. These datasets can be used to generate ETA forecasts that are extremely accurate when combined with machine learning models like LSTM[3].

According to recent studies, ETA accuracy is greatly increased by using contextual data, such as traffic density from APIs (Google Maps, TomTom), weather data from meteorological APIs, and local event schedules. However, such a multi-factor predictive technique is not often used in commercial implementations.

1.2 Research Gap

While GPS tracking for buses is common in many cities, most systems:

- Do not integrate **real-time traffic, weather, and event data** into predictions.
- Rely on rule-based or linear estimation models, which cannot capture complex temporal dependencies.

- Lack affordable **IoT-based architectures** that work in areas with inconsistent internet connectivity.
- Do not provide **open APIs** for third-party integration (e.g., Google Transit, city dashboards).

By creating an inexpensive, Internet of Things-based solution that integrates sensor data, GSM/Wi-Fi modules, and a cloud-hosted LSTM model for ETA prediction—optimized for developing regions—this study fills these gaps across various demographics and cultural settings. Researchers can demonstrate the technology's legitimacy and open the door for its broader use in clinical and other contexts by filling in these validation gaps[4].

1.3 Research Problem

Inaccurate bus arrival forecasts result in unhappy passengers, wasteful fleet use, and a decline in public confidence in transit systems. Conventional approaches are unable to manage[5]:

• Changing traffic circumstances

• Delays caused by the weather

• Unplanned incidents that impact routes

In order to provide passengers and transit authorities with precise arrival information, this project focusses on real-time bus tracking in conjunction with intelligent ETA prediction.

2.0 Objectives

2.1 Main Objective

The purpose of this platform is to increase the effectiveness and convenience of public bus services for both drivers and passengers. To guarantee more seamless travel

experiences, it offers schedule assistance, route management, and real-time information. Drivers gain from improved route guidance, less idle time, and improved cooperation with the transportation system as a whole, while passengers can readily access bus schedules, track bus whereabouts, and plan their trips with greater accuracy. Through facilitating communication between drivers and passengers, the platform improves both parties' efficiency, dependability, and satisfaction.

2.2 Specific Objectives

Performing this research has the following sub-objectives.

- Use an Arduino/ESP32 to put in place a GPS-based tracking system.
- Create a cloud data pipeline for processing and storing bus data in real time.
- Utilize GPS, traffic, weather, and event data to train and implement an LSTM model.

Connectivity modules for GSM and WiFi should be integrated for smooth data transfer.

- Provide a dashboard on the web or mobile device for administrators and passengers.

3.0 METHODOLOGY

3.1 System Overview

The proposed Smart Bus System is a modular and integrated solution designed to enhance public transportation in Sri Lanka through real-time transit optimization and improved passenger experience. The system consists of four key components, each addressing important aspects of urban bus operations by leveraging cutting-edge technologies such as IoT, computer vision, artificial intelligence, and human computer interaction.

The first component focuses on IoT-based real-time bus tracking and ETA prediction. It utilizes a hybrid connectivity strategy that combines GSM and Wi-Fi

to continuously transmit data in diverse environments. Advanced LSTM-based AI models trained on GPS, weather, traffic, and event data enable precise prediction of bus arrival times. Additionally, edge processing on Arduino hardware offers a low-cost, scalable solution suitable for deployment in developing regions. The second component employs vision-based passenger counting and occupancy monitoring by integrating computer vision and deep learning optimized for embedded systems without GPUs. This module combines binocular depth cameras with thermal sensors to ensure accurate passenger detection while preserving privacy through anonymized data processing.

The third component introduces a mobile-based passenger interaction and accessibility system, emphasizing inclusive design. It supports multi-modal interactions such as audio, vibration, and multilingual features to accommodate differently-abled users. Real-time input through QR codes or a mobile app enables dynamic adjustments to system behavior. Moreover, passengers receive crowd aware journey recommendations based on live seat availability and occupancy forecasts.

The fourth component is the central fleet management and route optimization engine, which applies data analytics, optimization algorithms, and fleet intelligence. It dynamically adjusts bus routes and schedules in response to passenger density and traffic, moving beyond static timetables. The AI-powered predictive scheduling with a feedback loop refines operations by learning from historical and current data. This engine also integrates train and ride-sharing services into a unified route planning system, facilitating seamless multimodal travel.

Together, these components create a comprehensive Smart Bus System aimed at improving operational efficiency, reducing passenger waits, enhancing accessibility, and promoting sustainable urban mobility. The modular design allows for parallel development and scalable deployment, addressing the multifaceted

challenges of public transit in Sri Lanka through innovative technology and user-centered solutions.

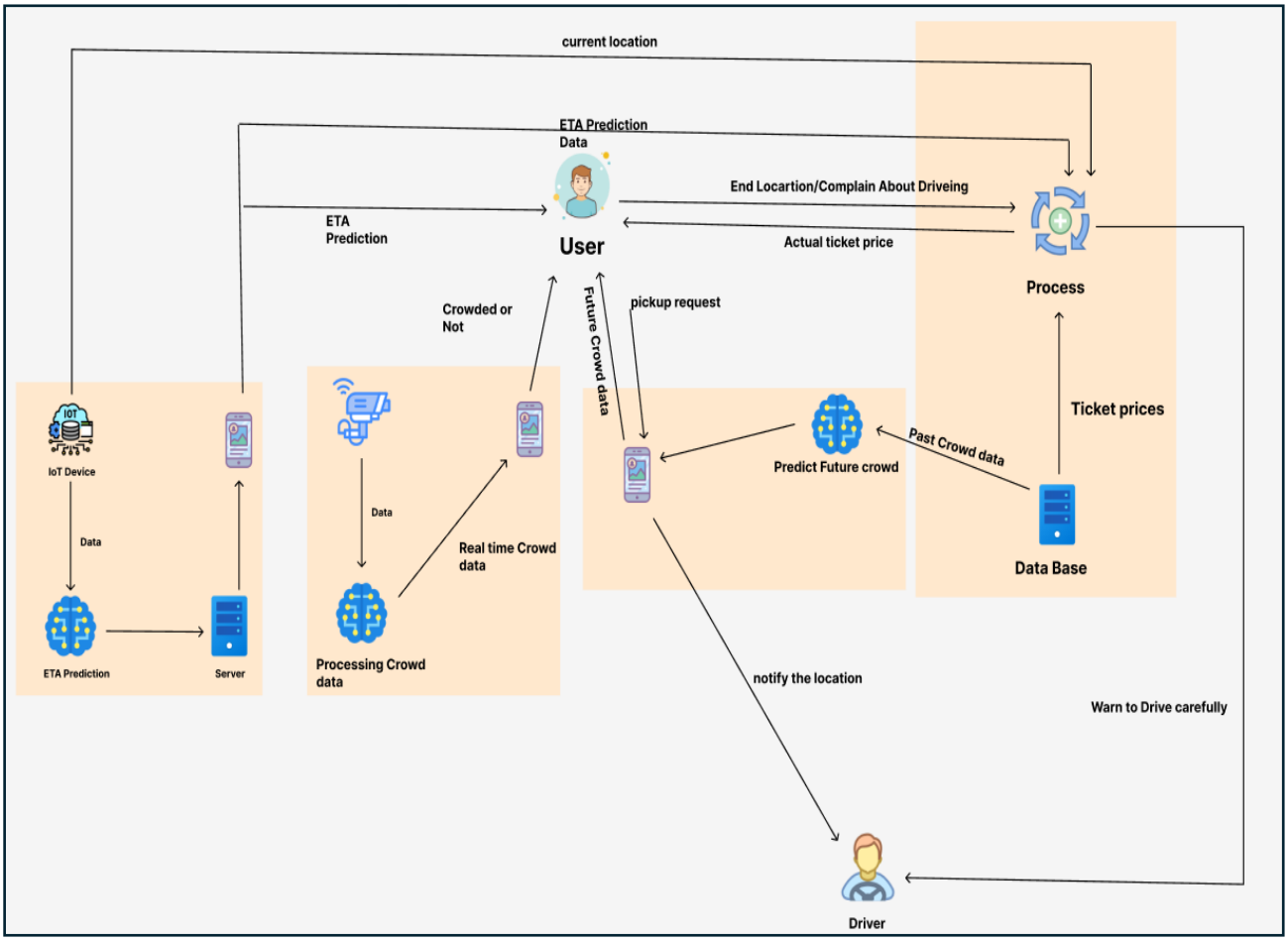

Figure 1: System Diagram

3.2 Individual Component

3.2.1 System Overview

The suggested system is an Internet of Things-based platform for real-time bus tracking and ETA prediction that aims to increase public transportation's dependability[1]. Every bus has an Arduino/ESP32 Internet of Things device that

is linked to a GPS and GSM/Wi-Fi module. These gadgets gather speed and location information and instantly transmit it to a cloud database.

The incoming GPS data is processed in the cloud by an LSTM (Long Short-Term Memory) machine learning model along with contextual information like local events, traffic, and weather. For forthcoming bus stops, the model provides precise Estimated Time of Arrival (ETA) predictions.

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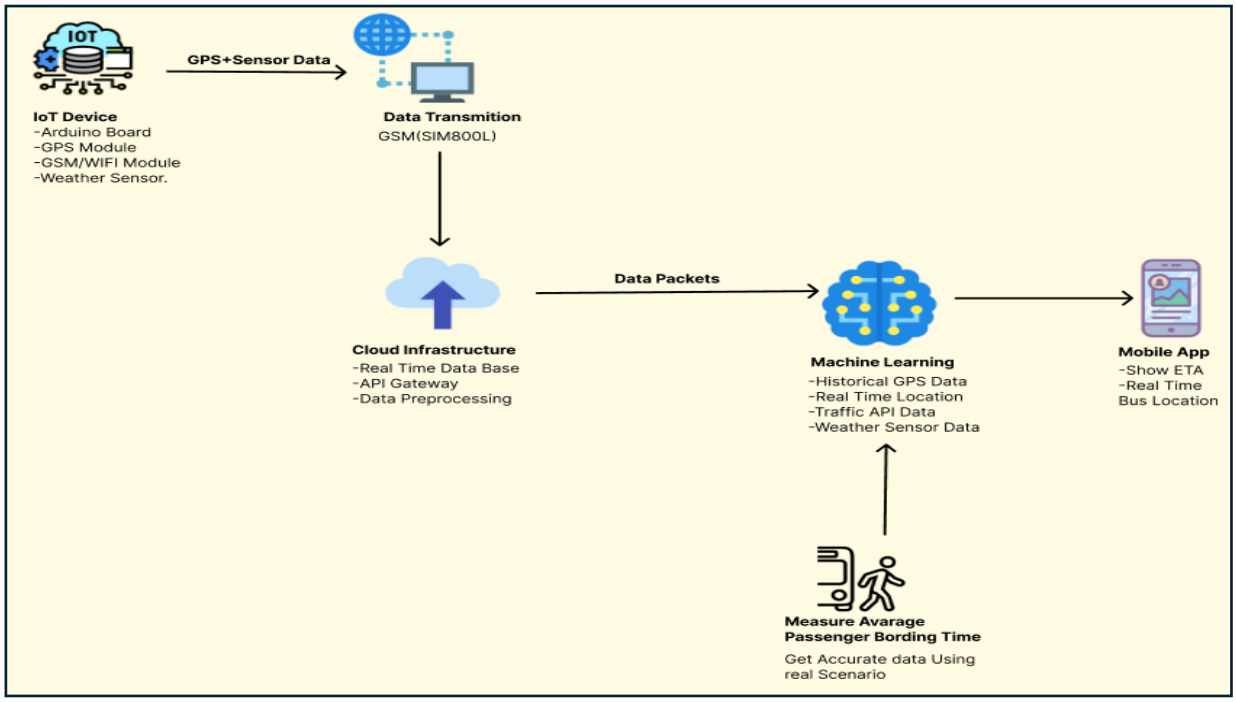


Figure 2: Individual Component System Diagram

3.3 Requirement Gathering

Requirement gathering will involve:

- Reviewing existing literature on GPS tracking, IoT communication, and ML-based ETA prediction[6].
- Studying public transport challenges in the Sri Lankan context. • Interviewing bus operators and passengers.

3.3.1 Past Research Analysis

In terms of previous research analysis, the subjects of bus tracking systems and ETA prediction are covered in a sizable number of research papers and publications. Nonetheless, a lot of articles took the ETA Prediction into consideration. Finding the methods and resources used to create all of the current tools and platforms was the primary goal of the previous study investigation. Additionally, it aided in identifying the issues that earlier researchers encountered.

3.4 Feasibility Study

3.4.1 Technical Feasibility

Technologies & Tools:

- Arduino/ESP32 microcontrollers
- GPS modules (Neo-6M)
- GSM (SIM800L) and Wi-Fi connectivity
- Cloud database (Firebase, AWS DynamoDB)
- Python-based LSTM model (TensorFlow/Keras)
- ReactJS for dashboard UI

3.4.2 Schedule Feasibility

This is an independent solution as a part of Sparkle Edu. This will be developed according to the timeline. Figure 6 and Table 2 illustrates the schedule management plan clearly.

3.4.3 Economic Feasibility

The system will use low-cost IoT hardware and scalable cloud resources with payasyou-go billing..

3.5 System Analysis

3.5.1 Software Solution Approach

1. Data Collection: Gather speed, timestamp, and real-time GPS coordinates from Internet of Things modules mounted on buses (Arduino/ESP32 with GPS & GSM/Wi-Fi modules). We will also gather other contextual data, like weather data (from meteorological sources), event data, and traffic data (via APIs).
2. Data Acquisition: Use Wi-Fi or GSM connectivity to send the collected data from IoT devices to a cloud-based database.
3. Preprocessing: Apply preprocessing techniques to the gathered data, including time-series alignment, handling missing values, clearing inaccurate GPS coordinates, and normalising input features (weather, traffic density, etc.).
4. Feature Extraction: From the preprocessed dataset, extract relevant features for ETA prediction. Some examples of features are:
    a. Current bus location (latitude, longitude)
    b. Distance to next stop
    c. Average bus speed
    d. Historical travel times on the same route

e. Real-time traffic congestion index

f. Weather impact factors (rain, temperature, etc.)

5. **Model Training:** Train an **LSTM (Long Short-Term Memory) neural network model** on historical and real-time data to predict ETA at bus stops. The LSTM is selected because of its ability to model sequential and timedependent data effectively.

6.**Model Evaluation:** Evaluate the performance of the model using standard metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Validation will be done using real-time test data from selected bus routes.

7.**System Deployment:** Deploy the trained LSTM model in the cloud, integrate it with the IoT data pipeline, and provide real-time ETA predictions to passengers via a web/mobile application.

3.5.2 Tools & Technology

- Arduino/ESP32 microcontrollers
- GPS modules (Neo-6M)
- GSM (SIM800L) and Wi-Fi connectivity
- Cloud database (Firebase, AWS DynamoDB)
- Python-based LSTM model (TensorFlow/Keras)
- ReactJS for dashboard UI

3.6 Project Requirements

3.6.1 Functional Requirements

- Real-time GPS tracking
- ETA calculation and display
- Multi-data integration (traffic, weather, events)

3.6.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested model are listed below.

- Usability
- Reliability
- Availability
- Accuracy
- Performance

3.7 Testing

It is essential to the effectiveness of this research when it comes to testing the findings. The first plan is to test the solution among a few cars in order to determine whether the suggested remedy works. We can determine if the device is functioning or not.

Nevertheless, this technique will be tested using the chosen bus services in the last phase of the study. This will function similarly to acceptance testing (beta, alpha). The majority of the testing will be done by hand.

Test Cases

Test Case ID	Description	Prediction	Test Steps	Expected Result
TC-1	Test GPS data capture from IoT device	GPS module connected to Arduino/ESP32 and powered on	1. Power on IoT device 2. Wait for GPS fix 3. Read latitude & longitude values	Correct GPS coordinates are displayed in the serial monitor / sent to cloud

TC-2	Test data transmission from IoT device to cloud	IoT device configured with GSM/Wi-Fi & SIM card	1. Capture GPS data 2. Send via GSM/Wi-Fi 3. Check cloud database for records	Data successfully stored in cloud database in real time
TC-3	Test cloud data storage reliability	Cloud database (Firebase/AWS) set up and connected	1. Send continuous GPS data for 30 mins 2. Check database logs 3. Verify no data loss	All data points are stored correctly without duplication or loss
TC-4	Test ETA prediction with LSTM model	Historical route data + trained LSTM model deployed	1. Input current GPS position, traffic & weather data	ETA is generated within acceptable error range (≤

			2. Run LSTM prediction 3. Retrieve ETA	2 min deviation)
TC-5	Test bus location display on mobile/web app	Mobile/web app connected to cloud backend	1. Open app 2. Select bus route 3. View live map	Real-time bus location displayed and updated every few seconds

TC-6	Test ETA display for selected bus stop	ETA prediction API integrated into app	1. Select a bus stop on the app 2. Request ETA 3. Compare with actual arrival	ETA shown to user with ≤10% error margin
TC-7	Test network failure handling	IoT device with GSM/Wi-Fi temporarily disconnected	1. Disconnect GSM/Wi-Fi 2. Capture GPS data locally 3. Reconnect network 4. Sync data	Data buffered locally and synced to cloud after reconnection

TC-8	Test system latency	IoT device connected & app running	1. Capture a GPS point 2. Measure time until it appears on app	End-to-end latency ≤ 5 seconds
TC-9	Test multibus tracking	3–5 buses with IoT devices connected	1. Track multiple buses 2. View on app 3. Verify each location	All buses tracked simultaneously with no overlaps or missing data
TC-10	Validate accuracy of	LSTM model trained and deployed	1. Compare predicted ETA with actual arrival	ETA error ≤ 2 minutes on average
	ETA predictions		at stop 2. Record error margin	

Table 1: Test Cases

3.8 Timeline

The proposed timeline for the project is as follows.

ID	Name	Start Date	End Date	Duration
1	Topic selection	16 May, 2025	26 May, 2025	11 Days
2	Topic selection completed	26 May, 2025	27 June, 2025	2 Days
3	TAF Assessment	30 June, 2025	16 July, 2025	17 Days
5	Charter submission	17 July, 2025	23 July, 2025	7 Days
6	Feasibility study	24 July,2025	30 July, 2025	7 Days
7	Literature review	31 July, 2025	5 Aug, 2025	6 Days
8	Project proposal draft	6 Aug, 2025	15 Aug, 2025	10 Days
9	Proposal presentation	16 Aug, 2025	9 Sep,2025	25 Days
10	Proposal presentation complete	10 Sep, 2025	10 Sep, 2025	0 Days
11	Proposal report	11 Sep, 2025	19 Sep, 2025	9 Days
12	Finalizing dataset	20 Sep, 2025	30 Sep, 2025	11 Days
13	Finalizing methodologies	1 Oct, 2025	5 Oct, 2025	5 Days
14	Data preprocessing	6 Oct, 2025	10 Oct,2025	5 Days
15	Text extraction	11 Oct, 2025	17 Oct, 2025	7 Days
16	Model building / device building	18 Oct, 2025	8 Nov, 2025	22 Days
17	Model building completed	9 Nov,2025	16 Nov, 2025	8 Days
18	Progress presentation 1	17 Nov, 2025	19 Nov, 2025	5 Days
19	Model building 2	25 Nov, 2025	27 Dec, 2025	32 Days
20	Model building 2 completed	4 Jan, 2025	7 Jan, 2025	3 Days
21	Back End completion	8 Jan, 2025	6 Feb, 2025	30 Days
22	Frontend completion	7 Feb, 2025	10 Mar, 2025	32 Days
23	First integration	11 Mar, 2025	28 Mar, 2025	18 Days
24	Progress presentation 2	29 Mar, 2025	31 Mar, 2025	3 Days
25	Progress presentation 2 deliverd	1 Apr. 2025	1 Apr, 2025	0 Days
26	Integration copletion	2 Apr, 2025	28 Apr, 2025	27 Days
27	Unit testing	29 Apr, 2025	10 May, 2025	12 Days
28	Interation testing	11 May, 2025	25 May, 2025	15 Days
29	Final document	26 May, 2025	15 Jun, 2025	21 Days
30	Final document completed	16 Jun, 2025	16 Jun, 2025	0 Days
31	Final presentation and VIVA	20 Jun, 2025	25 Jun, 2025	6 Days

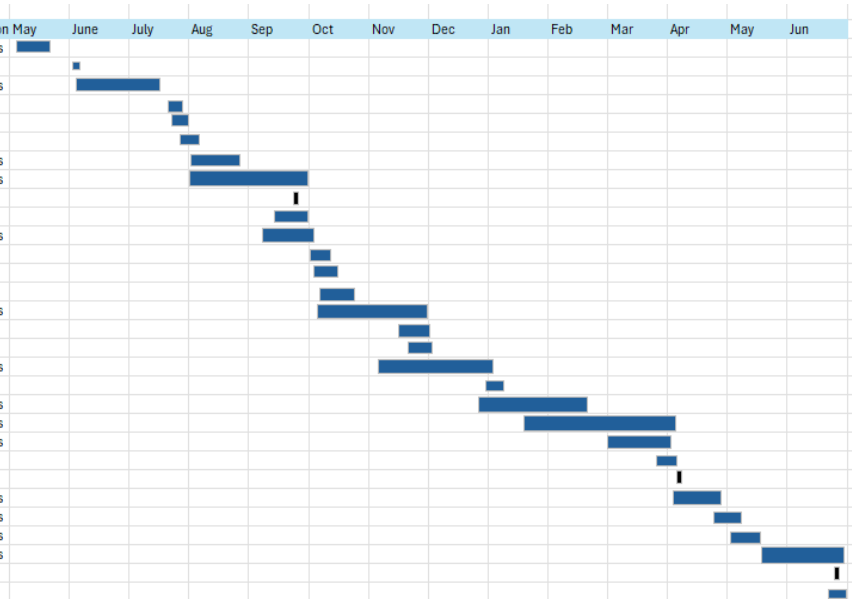


Figure 3: Gantt Chart for Project Schedule Management

3.9 Communication Management Plan

The Communications Management Plan ensures that the team, supervisor, and co-supervisor, among other project stakeholders, receive the information they require to perform their duties effectively throughout the project. For the project to be successful, communication activities need to be well planned and executed.

The most efficient and successful methods of communicating with different stakeholders are described in this strategy. The target audience, content, format, frequency, and expected outcomes of each communication channel are all described and documented. It also describes each stakeholder's responsibilities, task-assignment procedures, and communication strategies based on their level of interest, influence, and project expectations.

The key objectives of this communication plan are:

- To ensure timely and accurate flow of information among all stakeholders.
- To promote transparency and accountability within the project team.
- To facilitate feedback loops that help in improving project deliverables.

- To maintain alignment between project goals, supervisor guidance, and team execution.

3.9.1 Communication Objectives:

Proactive communication is important on all projects. Communication needs to be:

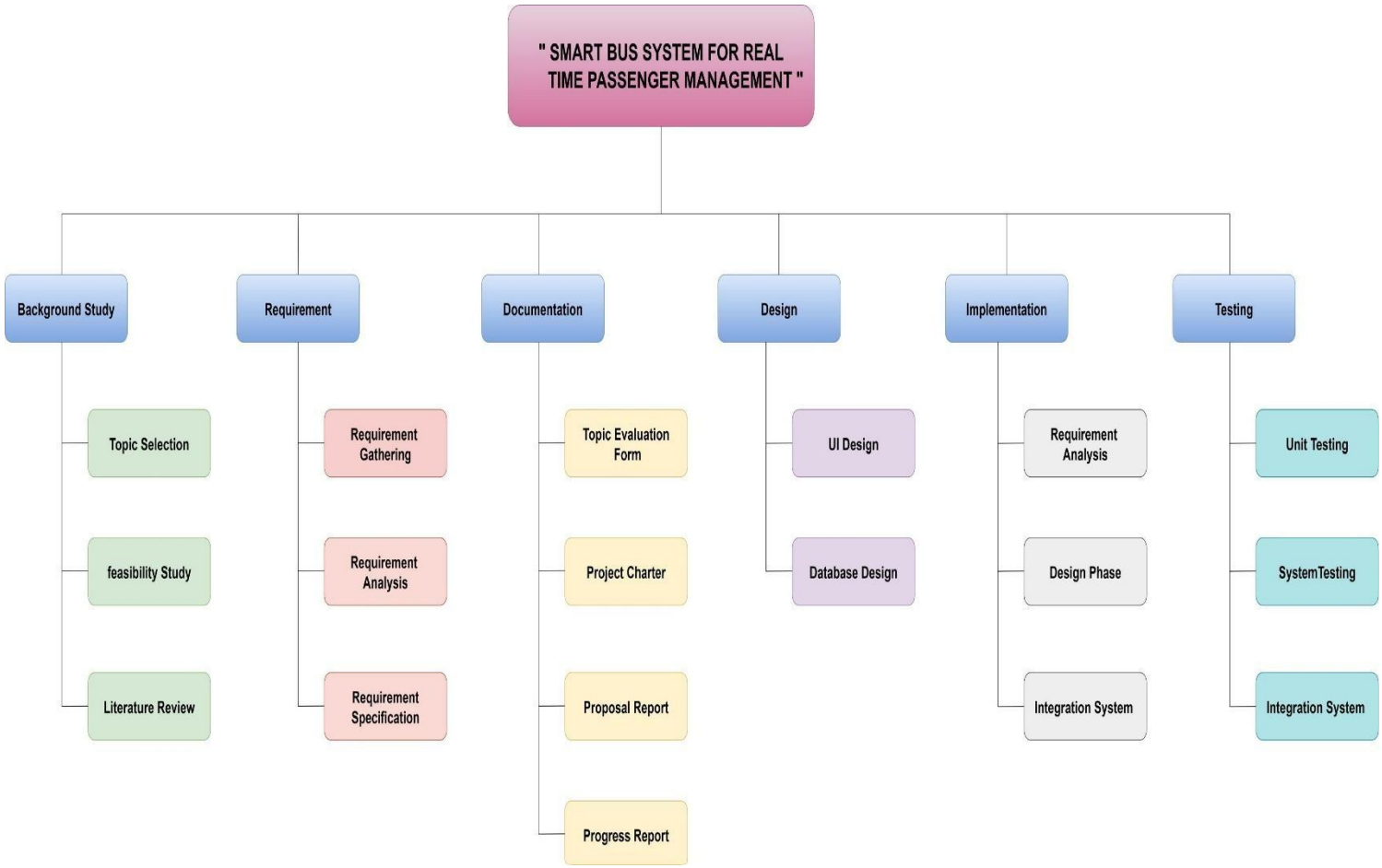
- Adequate: in the right format and right content.
- Specific: for the targeted audience.
- Sufficient: providing all the necessary information.
- Concise: brief, avoiding repetition and non-important information.
- Timely: addressing points at the right time.

3.9.2 Communication Media:

The communication media that will be used for the project are:

1. **Email** – For official communications, documentation, and approvals.
2. **Documents (MS Word / PowerPoint)** – For sharing project reports, proposals, and presentations.
3. **Instant Messaging (WhatsApp, MS Teams Chat)** – For day-to-day updates and quick clarifications.
4. **Meetings (Physical or MS Teams Virtual Meetings)** – For project discussions, milestone reviews, and progress presentations.
5. **Phone Calls** – For urgent communications requiring immediate response.

WORK BREAKDOWN



4.0 COMMERCIALIZATION

Both public and private bus companies can enhance their transportation offerings by utilizing the IoT-Based Real-Time Bus Tracking & ETA Prediction technology. Operators can more effectively manage their fleets, and passengers will benefit from knowing the precise bus location and arrival time. The system may be offered as a subscription service, allowing bus firms to receive real-time tracking and ETA predictions in exchange for a nominal price. Additionally, it can be included into a

passenger mobile app, which may eventually have premium services or ads. This technique is practical and affordable for large-scale application because the cloud system is scalable and the hardware is inexpensive. The system can eventually be expanded to include intercity coaches and school buses, increasing its commercialization potential.

5.0 BUDGET

Since the outcome of the proposed model is a software-based solution, there are no hardware components connected to the implementation. The primary source of the cost will be the subscription fees to the cloud provider for the computing power of the virtual machines. However, there will be some other costs expected to as be given in the table below.

Type	Cost
IoT hardware (Arduino/ESP32, GPS, GSM module, sensors) per bus	10000 LKR
Cloud hosting & database services (annual)	12100 LKR
Internet/data SIM charges(Aproximately) (per bus annually)	6000 LKR
Stationery, documentation & miscellaneous	3000LKR
TOTAL	***31100 LKR***

Table 2: Cost Management Plan

This amount may differ according many reasons.

6.0 SUMMARY

The goal of this project is to develop an affordable, Internet of Things-based real-time bus tracking and ETA prediction system specifically for Sri Lanka. Through the integration of GPS, cloud storage, IoT communication modules, and an LSTM prediction

model, the system will enhance operational efficiency, passenger happiness, and reliability.

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

Name	Student ID	Signature
Pathiraja H.P.M.O.N	IT22230492	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....

Signature of the supervisor:

(Mr. Nelum Amarasena) Date: 29/08/2025

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Signature of the co-supervisor:

(Mrs. Uthpala Samarakoon) Date: 29/08/2025

Abstract

Efficient fleet management and route optimization are critical to improving public transit systems. This component focuses on developing a Central Fleet Management and Route Optimization Engine that leverages data analytics and ML and AI-driven optimization to enhance real-time transit operations. The system introduces a demand-aware route adjustment mechanism that dynamically modifies routes based on passenger density and traffic conditions, moving beyond traditional static scheduling. Additionally, it implements a predictive scheduling framework with a feedback loop that refines transit schedules using historical and live usage data to improve service reliability. A cross-modal integration engine further enhances route planning by incorporating train and ride-share services into a unified platform, providing seamless transit experiences. This integrated approach aims to optimize fleet utilization, reduce passenger waiting times, and improve overall system efficiency, contributing significantly to smarter and more responsive urban transit solutions.

Keywords: Fleet Management, Route Optimization, Data Analytics, Predictive Scheduling, Multimodal Transit Integration

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List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
ETA	Estimated Time Arrival
LSTM	Long Short-Term Memory
GPS	Global Positioning System
SaaS	Software as a Service
API	Application Programming Interface
UI/UX	User Interface / User Experience
GSM	Global System for Mobile Communications
GPU	Graphics Processing Unit
AWS	Amazon Web Services
JWT	JSON Web Token
CI/CD	Continuous Integration / Continuous Deployment
IDE	Integrated Development Environment
ELK	Elasticsearch, Logstash, Kibana
HCI	Human-Computer Interaction
QoS	Quality of Service

Table 1: Table of Abbreviations

1.0 INTRODUCTION

1.1 Background and Literature

Many people in Sri Lanka rely on public transportation to get around for work, school, and daily living. Although the number of bus passengers in Sri Lanka is increasing, older bus systems frequently employ set routes and schedules that don't adjust to traffic in real time or the number of passengers waiting at certain stops. In populous cities like Colombo, this can lead to unneeded delays, crowded buses, and inconvenience for passengers [1], [2].

With new technology, there are smart solutions for these problems. Using data analytics and AI, bus fleets can plan routes more efficiently, make live adjustments based on real passenger data, and work together with other transport like trains or ride-sharing services [3], [4]. These smart systems collect information about the number of waiting passengers, traffic conditions, and even special events, which increases the route optimization and, provide a smoother journey for everyone [5].

Real-time data on traffic and passenger numbers can be used to improve the responsiveness and dependability of public transportation, according to studies conducted in other nations and pilot projects in Sri Lanka [5], [6], [7]. For instance, integration with other forms of transportation can provide passengers with more options and cut down on waiting times, while ML techniques can assist in forecasting the optimal schedules and avoiding traffic congestion [8]. However, most Sri Lanka's current systems still concentrate on predetermined procedures, and fully integrated, affordable solutions that can quickly adjust to local needs are required [1], [2], [7].

The proposed Central Fleet Management and Route Optimization Engine is designed to solve these issues for Sri Lankan bus systems. It automatically adjusts bus routes and schedules using live data and keeps learning from new information through AI feedback [9]. It also connects buses with train and ride-share services, giving passengers a better, more connected travel experience [10]

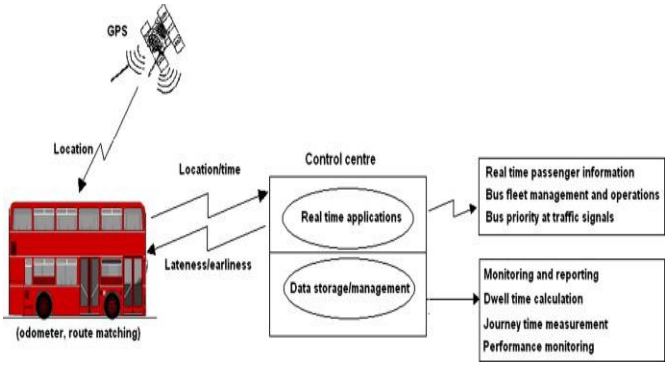

Figure 1: Central Fleet Management and Route Optimization
Source: Google Scholar

1.2 Research Gap

In Sri Lanka, most peoples' everyday lives depend heavily on public transportation, yet the country's bus systems are still mostly based on set routes and static schedules that don't adapt to changing traffic or passenger demand. In high-density cities like Colombo, this restriction leads to delays, crammed buses, and decreased passenger satisfaction. The potential of real-time, data-driven fleet optimization has been shown by international research and pilot programs, but Sri Lanka still lacks a fully functional, adaptable system that can react quickly to shifting travel conditions.

A significant deficiency in current literature and practice is the absence of demand-aware route adjustment. Real-time bus route and frequency adjustments based on passenger density, traffic jams, or special events are not possible with the majority of currently in use systems. Resources are either underutilized or overstretched in the absence of such flexibility, which results in inefficiencies and annoyance.

The lack of predictive scheduling tools with continuous learning represents another important research gap. Sri Lanka's current bus systems still use antiquated static timetables, despite research in other contexts emphasizing the use of AI and ML to improve scheduling. Predictive models that use both historical and real-time data in the Sri Lankan setting to increase accuracy and responsiveness have not received much attention.

Cross-modal integration is also still not well understood. The systems in place now almost entirely concentrate on buses, hardly ever coordinating with ridesharing, railroads, or last-mile services. A smooth public transportation system might cut down on waiting times and enhance the passenger experience overall, but this lack of integration limits that possibility.

There is a lack of affordable, scalable, and locally relevant alternatives, to sum up. Numerous implementations now in place are either imported models or small-scale pilots that fall short in taking into consideration Sri Lanka's distinct travel habits, infrastructure

limitations, and socioeconomic considerations. Adoption of smart fleet management will continue to be restricted unless these localization and scalability concerns are resolved.

By addressing these gaps demand aware route adjustment, predictive scheduling with feedback loops, and cross-modal integration the proposed Central Fleet Management and Route Optimization Engine aims to transform Sri Lanka's bus transport system into a more efficient, adaptive, and passenger-friendly service.

Product/ Platform	Demand-Aware Route Adjustment	Predictive Scheduling with Feedback Loop	Cross-Modal Integration (Bus + Train + Ride-share)	Real-Time Optimization
Google Maps Transit	NO	NO	Partial (Train/Bus only)	NO
Moovit	NO	NO	Partial (Multi-modal routing, but limited in Sri Lanka)	NO
SLTB Bus Scheduling System (Sri Lanka)	NO	NO	NO	NO
Ride-Sharing Platforms (e.g., PickMe, Uber)	NO (Focus on individual rides)	NO	NO (Isolated service)	YES (for ride allocation only)
Proposed Central Fleet Management & Route Optimization Engine	YES	YES	YES	YES

Figure 2: Competitive Analysis

1.3 Research Problem

In Sri Lanka, traditional bus systems confront significant challenges because of the country's fast urbanization and rising demand for effective public transit [1], [2]. Due to its inability to adjust to real-time variables like passenger density and road congestion, fixed bus schedules and routes frequently result in delays, overcrowding, and unsatisfactory commuter experiences [2], [4]. Furthermore, bus operations seldom interface with other forms of transportation, such as railroads or ride-sharing services, which restricts the effectiveness and convenience of travel [3], [4].

Currently available public transportation options in Sri Lanka are primarily based on static schedules that do not take into consideration real-time data, which results in inefficiencies and increased wait times for users [1], [5]. As a result, private vehicle use rises and public transportation becomes less appealing, which exacerbates traffic congestion and pollution [2]. Furthermore, many current systems do not have intelligent scheduling that can learn from and get better based on real-time and historical usage patterns [5], [8].

Therefore, this research focuses on developing a Central Fleet Management and Route Optimization Engine that utilizes demand-aware route adjustments, AI and ML -powered predictive scheduling with a feedback loop, and cross-modal integration with trains and ridesharing [4], [5], [9], [10]. By dynamically adjusting bus routes and schedules according to passenger demand and traffic conditions in real time, and by linking various transport modes, this system aims to enhance operational efficiency, reduce travel times, and improve passenger satisfaction across Sri Lanka’s public transit network [1], [2], [4].

2.0 OBJECTIVES

2.1 Main Objective

To develop a comprehensive, state-of-the-art Smart Bus System that leverages advanced technologies across multiple integrated components to optimize real-time public transit operations and significantly improve the passenger experience in Sri Lanka. This research aims to create an intelligent, scalable, and inclusive platform that addresses critical urban transportation challenges by combining IoT, computer vision, AI-driven analytics, and human-centered mobile interfaces. The system will provide seamless, data-driven solutions for real-time bus tracking and ETA prediction, accurate passenger counting and occupancy monitoring, interactive and accessible passenger engagement, and intelligent fleet management with dynamic route optimization. Ultimately, this project aspires to enhance transit efficiency, reduce congestion, promote sustainability, and foster accessibility for all commuters, contributing to a smarter and more resilient urban mobility ecosystem.

2.2 Specific Objectives

Develop a Smart Central Fleet Management and Route Optimization Engine that enhances public transport efficiency by dynamically adjusting bus routes and schedules based on real-time passenger demand and traffic conditions in Sri Lanka.

- Provide a demand-aware route adjustment system for live optimization of bus routes and schedules.
- Implement an AI and ML-based predictive scheduling model with a continuous feedback loop using historical and real-time transit data.
- Incorporate a cross-modal integration engine connecting bus services with trains and ridesharing to enable seamless multi-modal trip planning.
- Enhance fleet intelligence by enabling real-time monitoring and management of bus operations.

- Improve the passenger experience by reducing waiting times, avoiding overcrowding, and increasing reliability.
- Design the system specifically to address the unique challenges and context of Sri Lanka's public transportation network.

3.0 METHODOLOGY

3.1 System Overview

The proposed Smart Bus System is a modular and integrated solution designed to enhance public transportation in Sri Lanka through real-time transit optimization and improved passenger experience. The system consists of four key components, each addressing important aspects of urban bus operations by leveraging cutting-edge technologies such as IoT, computer vision, artificial intelligence, and human-computer interaction.

The first component focuses on IoT-based real-time bus tracking and ETA prediction. It utilizes a hybrid connectivity strategy that combines GSM and Wi-Fi to continuously transmit data in diverse environments. Advanced LSTM-based AI models trained on GPS, weather, traffic, and event data enable precise prediction of bus arrival times. Additionally, edge processing on Arduino hardware offers a low-cost, scalable solution suitable for deployment in developing regions.

The second component employs vision-based passenger counting and occupancy monitoring by integrating computer vision and deep learning optimized for embedded systems without GPUs. This module combines binocular depth cameras with thermal sensors to ensure accurate passenger detection while preserving privacy through anonymized data processing.

The third component introduces a mobile-based passenger interaction and accessibility system, emphasizing inclusive design. It supports multi-modal interactions such as audio, vibration, and multilingual features to accommodate differently abled users. Real-time input through QR codes or a mobile app enables dynamic adjustments to system behavior. Moreover, passengers receive crowd-aware journey recommendations based on live seat availability and occupancy forecasts.

The fourth component is the central fleet management and route optimization engine, which applies data analytics, optimization algorithms, and fleet intelligence. It dynamically adjusts bus routes and schedules in response to passenger density and traffic,

moving beyond static timetables. The AI-powered predictive scheduling with a feedback loop refines operations by learning from historical and current data. This engine also integrates train and ride-sharing services into a unified route planning system, facilitating seamless multimodal travel.

Together, these components create a comprehensive Smart Bus System aimed at improving operational efficiency, reducing passenger waits, enhancing accessibility, and promoting sustainable urban mobility. The modular design allows for parallel development and scalable deployment, addressing the multifaceted challenges of public transit in Sri Lanka through innovative technology and user-centered solutions.

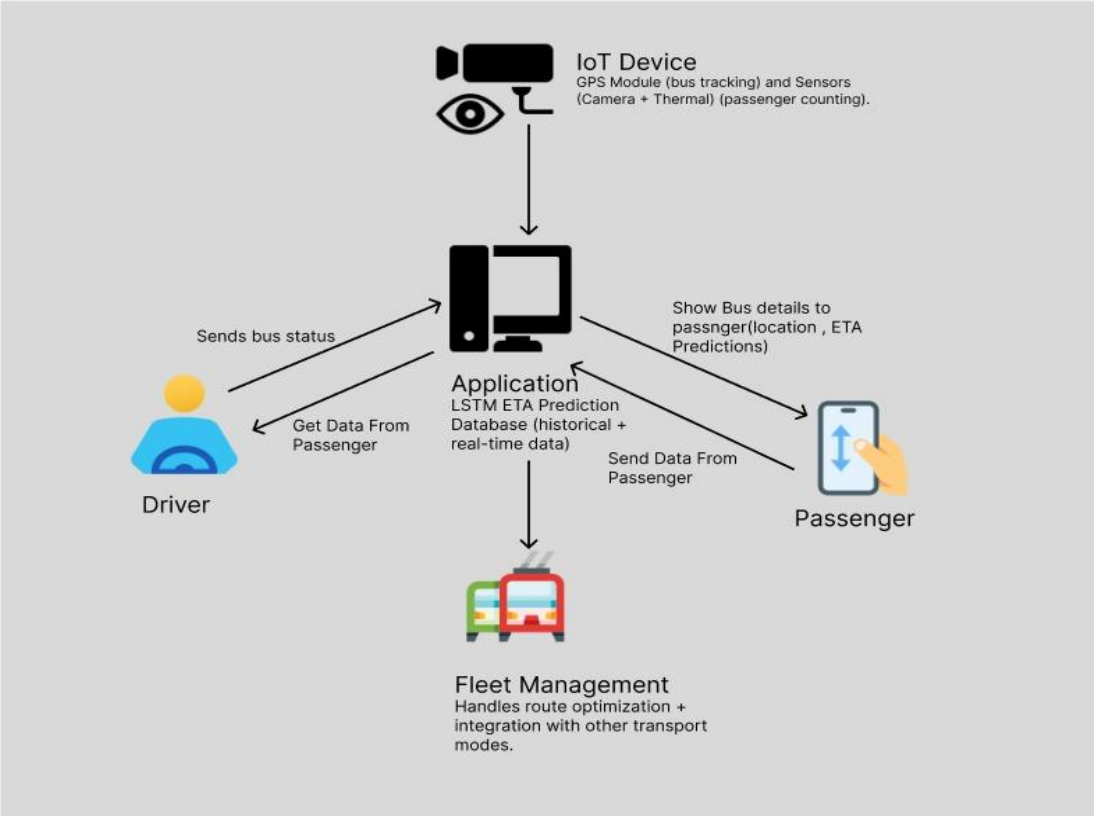

Figure 3: System Overview

3.1.1 Individual Component

Central Fleet Management and Route Optimization Engine, serving as the analytical and control core of the system, this engine employs data analytics, optimization algorithms, and fleet intelligence to adapt bus routes and schedules dynamically based on live passenger density and traffic data. It includes an AI-driven predictive scheduling system with a feedback loop refining operations from historical and real-time inputs. A cross-modal integration engine connects bus services with trains and ride-share platforms, enabling seamless multi-modal route planning for enhanced transit efficiency.

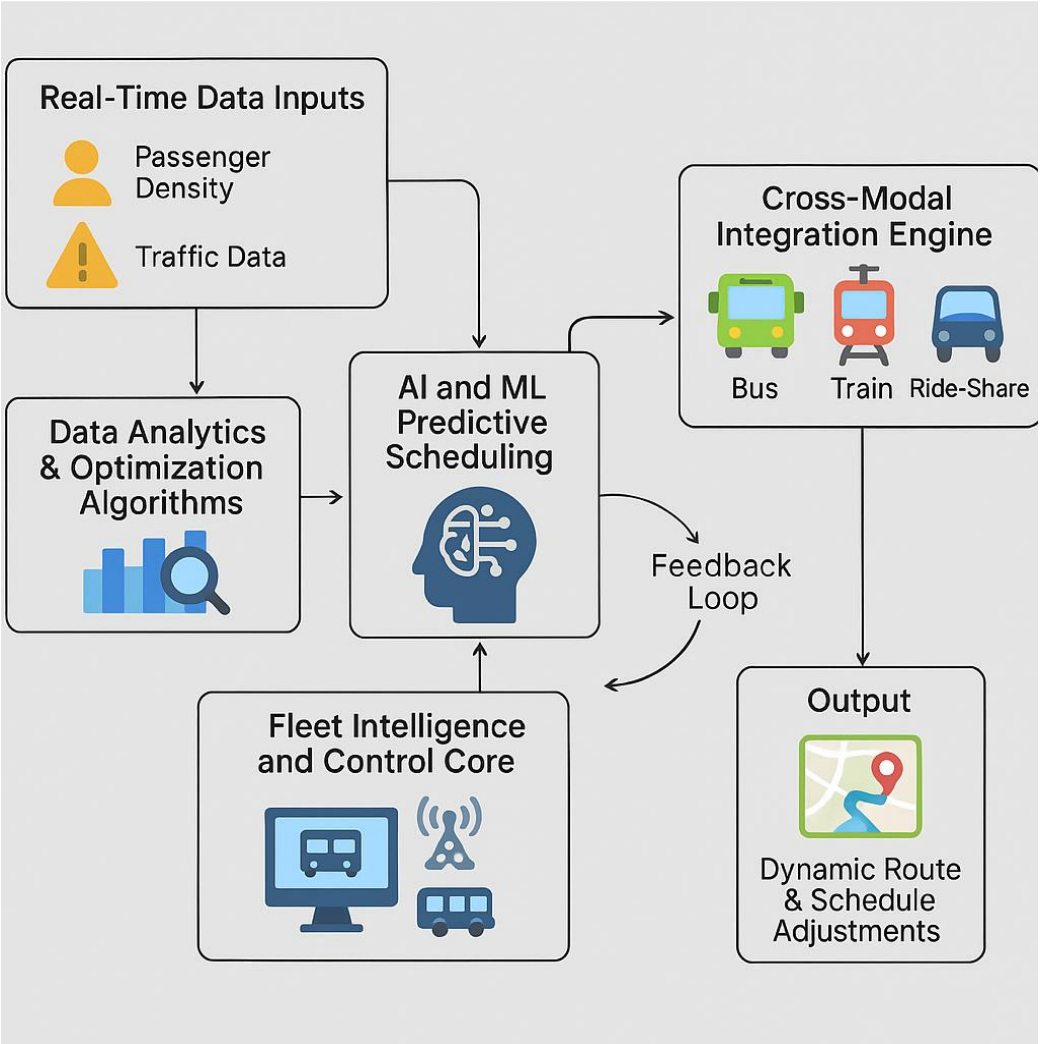

Figure 4: Individual Component Overview

3.2 Requirements Analysis

Requirement gathering for Component 4: Central Fleet Management and Route Optimization Engine was conducted through a comprehensive study involving several approaches. This included an extensive review and analysis of past research and advancements in public transit optimization, data analytics, and AI-driven fleet management. Additionally, existing transportation management systems and solutions were critically evaluated to identify gaps and areas for improvement. A wide range of online resources, including academic articles, technical reports, and industry whitepapers, were examined to understand current trends and best practices. Real-world transportation scenarios and challenges faced by urban transit authorities, particularly in Sri Lanka, were also considered to ensure the component addresses practical needs effectively. This multi-faceted requirement gathering process provided a solid foundation for defining the objectives, functionalities, and constraints of the component.

3.2.1 Past Research Analysis

A significant amount of prior research on Central Fleet Management and Route Optimization has been devoted to the application of data analytics, AI, ML and optimization methods to enhance public transportation systems. Predictive scheduling, dynamic routing, multi-modal transport integration, and real-time passenger demand forecasts are among the main areas of interest. The application of machine learning models, such as LSTM and reinforcement learning, to demand forecasting and fleet management has been the subject of numerous studies.

3.3 Feasibility Study

3.3.1 Technical Feasibility

3.3.1.1 Knowledge of Technologies

To develop the Central Fleet Management and Route Optimization Engine knowledge in the following technologies and methods is essential:

- Data Analytics and Big Data Processing: Processing large amounts of historical and real-time transit data, including passenger density and traffic conditions, is essential to this, necessitating proficiency with scalable data analytics tools.
- AI and ML: AI techniques, particularly predictive modeling, are used to forecast passenger demand and traffic patterns. Skills in machine learning algorithms like Long Short-Term Memory (LSTM) networks and reinforcement learning are crucial for adaptive scheduling and route optimization.
- Optimization Algorithms: Knowledge of optimization techniques, such as genetic algorithms, linear programming, or methods, is necessary for dynamically adjusting routes and schedules to improve fleet efficiency.
- Cloud Computing and Edge Processing: Understanding cloud infrastructure for scalable data storage and computation is required, alongside edge computing concepts to enable low-latency processing at the network edge for timely decision-making.
- Integration and API Development: Familiarity with designing and implementing APIs is needed for integrating multiple transportation modes (bus, train, ride-share like PickMe, Uber) and connecting with external traffic management systems.
- Fleet Intelligence Systems: Concepts related to fleet management, including vehicle tracking, status monitoring, and operational control protocols, form the basis for intelligent fleet coordination.
- Real-Time Systems and Event Processing: Experience with event-driven architecture and real-time data streaming platforms ensures the system can respond quickly to changing conditions.

- Security and Data Privacy: Knowledge of cybersecurity principles and privacy-preserving techniques is important to safeguard operational and passenger information.
- Human-Computer Interaction (HCI): Understanding how operators interact with fleet management dashboards aids in designing usable and effective control interfaces.
- Cross-Modal Transport Planning: Expertise in multi-modal route planning algorithms is essential to unify scheduling across different transit services for seamless passenger journeys.

3.3.1.2 Knowledge of Tools

To create the Central Fleet Management and Route Optimization Engine, each team member should be well-versed in the project management software, development tools, and frameworks that are necessary for successful and efficient implementation.

3.3.1.3 Data collection Knowledge

Data for this component can be collected through multiple means including:

- On-bus real-time sensors and gadgets (such as GPS and IoT passenger counts) that provide real-time information on the position and occupancy of buses.
- Real-time information on traffic congestion is provided by traffic monitoring systems.
- Transport authorities retain historical transit data records.
- User information on preferred routes and demand-responsive inputs is gathered using mobile apps.
- External data sources include ride-sharing service APIs (e.g., Uber, PickMe, etc.).
- Surveys and interviews with passengers and fleet operators to obtain qualitative information.

3.3.2 Schedule Feasibility

It functions as a separate but essential component of the larger Smart Bus System, which is intended for Real-Time Passenger Management and Transit Optimization. This component's development will follow a thorough schedule shown in the project's Gantt chart (Figure 6), guaranteeing timely delivery and alignment with other system modules.

3.3.2 Economic Feasibility

The Central Fleet Management and Route Optimization Engine is a cost-effective solution that fits into the larger architecture of the Smart Bus System. Implementation costs are substantially lower than those of many similar systems since the system makes use of cloud computing resources and pre-existing infrastructure to reduce the requirement for additional hardware investments.

The main expenditures are subscriptions to cloud services for processing data and computing power, as well as running costs for things like stationery, publications, and internet use. While guaranteeing scalability and robustness, these expenses are comparatively minimal.

The overall cost, however, might change based on changes in the economy and other outside variables that impact resource prices. Budget section includes a tentative cost estimate for transparent financial planning and management.

3.4 System Analysis

3.4.1 Software Solution Approach

The following outlines the approach for developing the Central Fleet Management and Route Optimization Engine:

- Data Collection: Collect data from multiple sources including real-time sensor feeds (passenger density using IoT device, bus GPS), traffic monitoring systems, historical transit records, and user inputs via mobile applications
- Data Integration and Preprocessing: Bring together different types of data, fix mistakes or mismatches, make sure time-based data lines up correctly, and pick out the useful details needed for analysis.
- Feature Extraction and Transformation: Derive meaningful variables such as passenger load patterns, average travel times, and congestion metrics essential for optimization.
- Demand Forecasting: Use AI and ML models such as LSTM to predict future passenger demand and traffic conditions based on historical and real-time data.
- Route and Schedule Optimization: Apply optimization algorithms to generate dynamic adjustments to bus routes and schedules that minimize delays and maximize fleet efficiency
- Cross-Modal Coordination: Integrate data and scheduling across different transport modalities (bus, train, ride-share) for unified trip planning.
- Model Training and Continuous Learning: Train predictive and optimization models continuously with new incoming data, applying feedback loops to improve accuracy and performance.
- Model Evaluation and Performance Monitoring: Utilize performance measures to assess the efficacy of the solution and direct improvements, such as system responsiveness, passenger wait times, and schedule adherence.

3.4.2 Tools & Technology

- Programming Languages - Python, JavaScript
- Frameworks and Libraries –
    - TensorFlow, Keras - for AI and ML model development
    - PyTorch - ML framework)
    - Scikit-learn - ML utilities)
    - Pandas, NumPy - data manipulation)
    - Apache Spark - big data processing if needed)
    - OpenAPI / Swagger - for API)
- Database Systems –
    - MongoDB: NoSQL for flexibility in data types including sensor and scheduling data.
    - Firebase: for real-time data streaming and integration.
- Cloud Platforms:
    - AWS / Google Cloud Platform / Microsoft Azure - for cloud computing, storage
    - Docker - containerization
    - Kubernetes - container orchestration
- Web Development Frameworks:
    - ReactJS + Vite
    - Node.js / Express
    - Tailwind CSS
- UI/UX Design Tools:
    - Figma / Draw.io - designing user interfaces, wireframes, prototypes
- Integrated Development Environment (IDE): Visual Studio Code
- Version Control: Git, GitHub
- Testing Tools:
    - Pytest - unit testing for Python code
    - Postman

    - Selenium - testing for UI components
- CI/CD:
    - Jenkins / GitHub Actions
- Configuration and Automation:
    - Ansible / Chef - configuration management
    - Terraform - infrastructure as code
- Monitoring and Logging:
    - Prometheus + Grafana - system and application monitoring dashboards
    - ELK Stack (Elasticsearch, Logstash, Kibana) for log analysis
- Collaboration and Project Management:
    - Microsoft Teams / Slack - communication
    - Jira / Trello - task and project management
- Security Tools:
    - SonarQube - code quality and security scanning
    - OAuth / JWT - authentication and authorization frameworks

3.5 Communication Management Plan

The Communications Management Plan assists in making sure that the supervisor, co-supervisor, and team members have the information they require to carry out their responsibilities during the project. Communication planning and execution are critical to the success of a project.

The communications management plan determines the most effective and efficient ways to communicate with various stakeholders. The document describes and records the target audience, the content, format, frequency, and expected results of the communication pieces. Additionally, it outlines the function of each stakeholder in the project, how to delegate duties to them, and the communication plan for each one according to their expectations, interests, and impact on the project.

3.5.1 Communication Objectives

- Adequate: right format, relevant content.
- Specific: tailored to stakeholders.
- Sufficient: covers progress, risks, updates.
- Concise: clear, focused, no overload.
- Timely: aligned with project milestones.

3.5.2 Communication Media

The communication media that will be used for the project are:

- Email
- Document (MS Word and/ PowerPoint)
- Phone call
- Meetings (using meeting rooms, conference phones, MS Teams)
- Chats (WhatsApp)

3.6 Timeline

The proposed timeline for completing the entire research, as shown in this Gantt chart notation, is within one year.

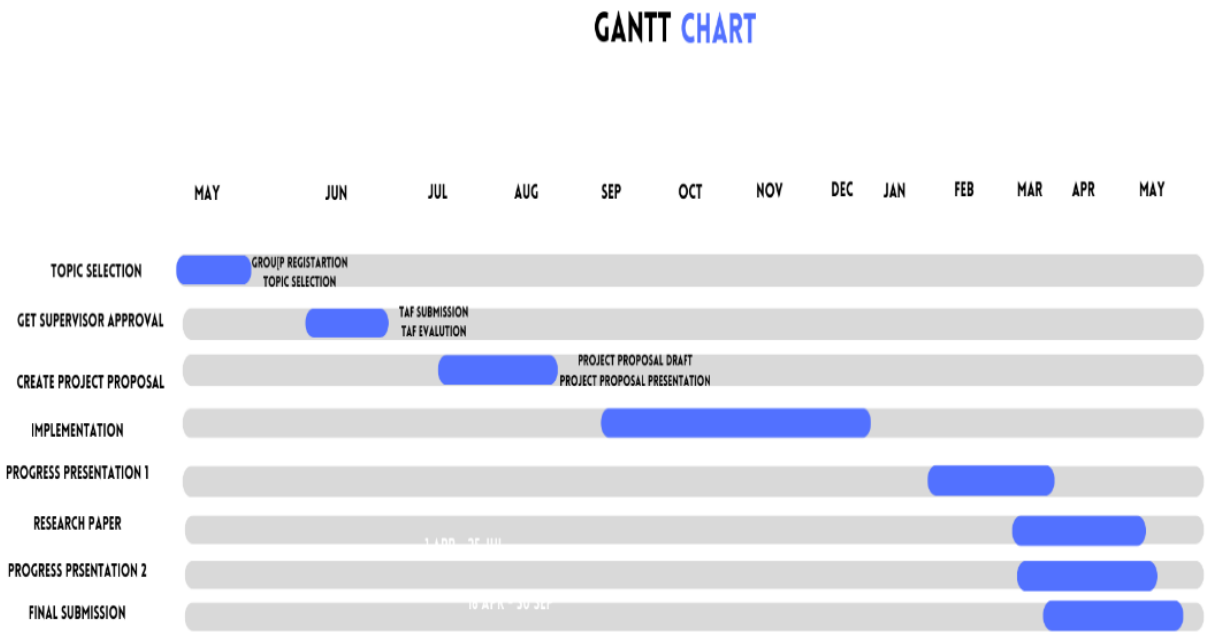

Figure 5: Gantt Chart

4.0 WORK BREAK DOWN STRUCTURE

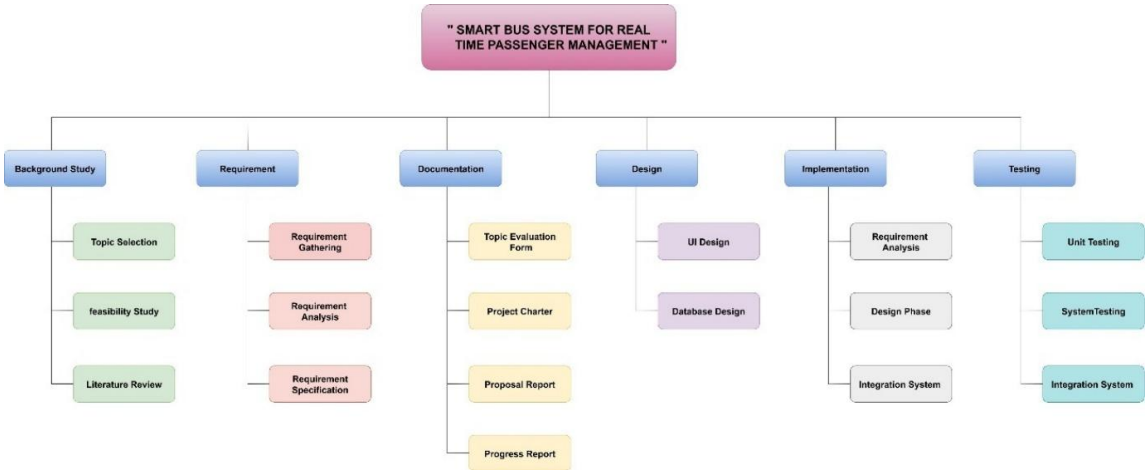

Figure 6: Work breakdown Structure

5.0 PROJECT REQUIREMENTS

5.1 Functional Requirements

Functional Requirements for Central Fleet Management and Route Optimization Engine

- Real-Time Data Collection:
  - Continuously collect passenger density data from buses and bus stops.
  - Gather live traffic condition data through integration with city traffic management systems or sensors.
  - Receive real-time bus location and operational status updates.
- Predictive Scheduling with Feedback Loop:
  - Utilize AI and ML algorithms to predict future transit demand and traffic patterns based on historical and live data.
  - Continuously refine and update bus schedules using feedback from past operations and current conditions.
- Cross-Modal Integration:
  - Facilitate integration of bus scheduling and routing with other transport modes such as trains and ride-share services.
  - Enable unified, multi-modal route planning accessible to fleet managers and passengers.
- Fleet Operational Management:
  - Provide a control interface for transit authorities to monitor and manage fleet operations in real time.
  - Provide a control interface for transit authorities to monitor and manage fleet operations in real time.
  - Support decision-making related to dispatching, resource allocation, and schedule adjustments.
- Data Storage and Reporting:
  - Maintain a database of historical transit data including passenger loads, trip times, and route performance.

    - Generate reports summarizing fleet performance metrics and trends for continuous improvement.
- User and System Security:
    - Ensure secure access control for fleet management interfaces and data communication.
    - Protect passenger and operational data confidentiality abiding by relevant privacy regulations.

5.2 Non-Functional Requirements

Non-Functional Requirements for Component 4: Central Fleet Management and Route Optimization Engine

- Performance:
    - The system shall process real-time data inputs and provide route and schedule updates with minimal latency to ensure timely decision-making.
    - It shall support simultaneous monitoring and optimization of the entire fleet without significant performance degradation.
- Scalability:
    - The engine must scale to handle increasing numbers of buses, passengers, and data sources as the transit network expands.
    - It should accommodate future integration with additional transport modes and new data streams.
- Reliability and Availability:
    - The system shall ensure high availability, targeting near 24/7 uptime to support continuous fleet operations.
    - It must implement failover mechanisms and backup systems to maintain service during outages or hardware failures.
- Security:
    - The platform shall enforce strict access controls to protect sensitive operational and passenger data.

    - Data transmission and storage must comply with encryption standards and privacy regulations.
- Usability:
    - Interfaces provided to fleet managers must be intuitive, providing clear visualization, alerts, and control options.
    - The system should support multiple user roles with appropriately tailored access and functionality.
- Maintainability:
    - The codebase and system architecture must facilitate easy updates, bug fixes, and enhancements.
    - Logging and diagnostic tools should be included to assist in troubleshooting and system monitoring.
- Interoperability:
    - The system shall support seamless integration with external data sources, traffic management systems, and other transportation modalities including trains and ride-share platforms (Uber, PickMe).
    - It should follow standardized communication protocols and data formats.
- Efficiency:
    - Resource consumption, including computational and network usage, should be optimized to minimize operational costs.
    - Processing algorithms should balance accuracy and speed to deliver effective optimization without excessive overhead.

5.3. User Requirements

- Real-time data regarding the number of people waiting at bus stops and aboard buses must be available to the system.
- The system should use sensors or city traffic services to gather information on the state of traffic now.
- Users must be able to see live bus locations and know if buses are arriving early or late
- The system should continuously update bus timetables, enhancing forecasts using historical and present data.
- Users must be able to plan trips that combine buses with other transport options like trains and ride-share services, all in one place
- Fleet managers require a basic control panel to monitor buses, get problem alerts, and alter schedules or routes as needed.
- To provide reports that aid in enhancing future services, the system ought to store Only authorized users should be able to make modifications or view sensitive data on the system, historical travel and passenger data.

5.4. Expected Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-01	Verify system collects real-time passenger density data	Fleet and sensor data streaming active	1. Start data collection from sensors 2. Monitor data updates	System displays accurate live passenger density across stops
TC-02	Verify predictive	Historical and real-time data available	1. Load historical data	Schedules update proactively

	scheduling updates		2. Trigger schedule update process	reflecting predicted demand
TC-03	Verify cross-modal trip planning	Bus, train, and ride-share schedules accessible	1. Input multi-modal trip request 2. View planned journey	System returns unified route combining bus, train, and ride-share
TC-04	Verify real-time fleet monitoring dashboard	Fleet operational and data feeds active	1. Access fleet dashboard 2. Check live status of buses and alerts	Dashboard shows up-to-date bus locations and performance metrics
TC-05	Verify alert generation for delays	Bus delay event simulated	1. Simulate bus delay 2. Check alert notification	System promptly alerts operators about the delay
TC-06	Verify manual route override functionality	AI-based route adjustment active	1. Operator manually changes route 2. Save and apply new route	Route changes take effect immediately overriding AI recommendations
TC-07	Verify data storage and report generation	Historical trips and passenger data present	1. Request performance report 2. Review generated report	System provides accurate, comprehensive fleet operation report
TC-08	Verify system security through user authentication	User account setup and permissions defined	1. Attempt login with valid/invalid credentials 2. Access restricted functions	Only authorized users can log in and access protected features
TC-09	Verify system scalability with increased fleet size	Simulated data for expanded fleet and network	1. Ramp up bus count and data streams 2. Monitor system responsiveness and stability	System maintains performance and reliability under high load
TC-10	Verify system adapts to	Active fleet operation and	1. Simulate unexpected	The system dynamically re-

	sudden traffic changes	real-time traffic data available	traffic congestion on a major route 2. Monitor route adjustments and schedule updates	routes affected buses and updates schedules to minimize delays

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Table 2: Expected Test Cases Table

6.0 COMMERCIALIZATION

Central Fleet Management and Route Optimization Engine designed primarily for public transportation authorities, fleet operators, and urban mobility planners. This system addresses the pressing challenges of inefficient bus operations and route planning in cities like Colombo, Sri Lanka, by using live passenger density, traffic data, and advanced AI-driven predictive analytics.

The target users for this solution include municipal transit agencies, private bus fleet operators, and multi-modal transport integrators. The component's ability to dynamically adjust routes and schedules in real time, along with seamless integration of trains and ride-share services, represents a significant advancement over traditional static planning systems currently in use.

This can be commercialized as a scalable software-as-a-service (SaaS) platform accessible via cloud infrastructure, offering subscription tiers based on fleet size and integration complexity. High-tier subscriptions may include advanced AI features such as predictive scheduling with feedback loops and cross-modal transport optimization, which are unique offerings in the urban transport management domain.

Partnerships with local government transit authorities, ride-sharing companies, and rail operators will be essential for data sharing and service integration, ensuring widespread adoption. Additionally, the system's modular architecture allows tailored deployment for different city scales and transport modes, increasing market flexibility.

By offering a smart, data-driven, and adaptable fleet management solution, this component can contribute significantly to improving public transit efficiency, reducing traffic congestion, and enhancing passenger satisfaction, positioning it competitively in the growing smart transportation industry.

7.0 BUDGET

Central Fleet Management and Route Optimization Engine is primarily a software-based solution, most of the costs will revolve around cloud computing, software development, and related services. There are no direct hardware costs associated with this component, as data collection relies on existing infrastructures and integrations.

The major cost driver will be subscription fees to cloud providers for virtual machine computing power, data storage, and analytics services. Other expected expenses include software licensing, internet usage, and miscellaneous operational costs.

Type	Cost (LKR)
Cloud computing and storage	18,000
Internet use and web hosting	5,000
Software licenses and tools	7,500
Publication and documentation	12,000
Stationery and miscellaneous	5,500
TOTAL	**48,000**

Table 3: Cost Management Plan

8.0 SUMMARY

The Central Fleet Management and Route Optimization Engine use cutting-edge optimization algorithms and real-time data analytics to dynamically improve public transportation operations. Based on passenger density, traffic, and historical data, this study continuously improves bus routes and schedules by utilizing AI-based predictive scheduling in conjunction with a feedback loop. Additionally, the system incorporates ride-sharing services and trains into unified route planning, allowing for smooth travel experiences.

The main objective of this is to give fleet operators and transit authorities an intelligent platform that enhances operational effectiveness, lessens traffic, and dynamically adjusts to shifting transit demands. This will help make public transit in cities more dependable, adaptable, and sustainable. The findings of this study have applications in improving the public transportation system in Sri Lanka and other developing nations, enabling more intelligent and adaptable urban mobility solutions.

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SMART BUS SYSTEM FOR REAL-TIME PASSENGER MANAGEMENT

(Real Time Crowd Calculation & Seat Reservation)

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Project Proposal Report

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Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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Project Proposal Report

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Mr. Nelum Amarasena

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

Public bus transportation serves as a vital mobility solution for millions of daily travelers in Sri Lanka, yet the sector remains challenged by late arrivals, overcrowding, limited accessibility for differently abled passengers, and inefficient fleet management. This research presents a modular Smart Bus System designed to address these challenges through the integration of advanced IoT, data analytics, computer vision, and inclusive mobile technologies. The proposed system comprises four core components: IoT-Based Real-Time Bus Tracking & ETA Prediction utilizing hybrid GSM and Wi-Fi connectivity with LSTM models for highly accurate arrival estimates; Vision-Based Passenger Counting and Occupancy Monitoring leveraging embedded deep learning and a hybrid sensor approach for anonymous, real-time crowd estimation; Mobile-Based Passenger Interaction and Accessibility System offering multi-modal user interfaces and demand-responsive routing to enhance inclusivity and communication; and a Central Fleet Management and Route Optimization Engine that employs dynamic, demand-aware route adjustments, predictive AI scheduling, and cross-modal trip planning with trains and ride-sharing services. Methodologically, our approach integrates real-time and historical transport data from sensors, GPS, user interactions, and traffic feeds into a unified analytical engine, enabling adaptive optimization and continuous schedule refinement. Anticipated outcomes include improved punctuality, optimized seat occupancy, enhanced passenger experience, and smarter resource use across multimodal transit networks. The study concludes with recommendations for scalable deployment in developing regions and outlines the expected impact on urban mobility, commuter satisfaction, and system-wide efficiency.

Keywords: Public Transport, IoT, Fleet Management, Data Analytics, Route Optimization, Inclusive Design, Urban Mobility

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Abbreviation

IT	Information Technology
GSM	Global System For Mobile Communication
Wi-Fi	Wireless Fidelity
LSTM	Long Short-Term Memory
AI	Artificial Intelligence
GPS	Global Positioning System
TC	Test Cases

1.0 INTRODUCTION

1.1 Background and Literature

Urban commuters often face uncertainty when using public transport, as they cannot determine whether an approaching bus is already full, partially occupied, or has available seating. This results in long waiting times, discomfort, and, in many cases, unsafe overcrowded conditions. At the same time, authorities such as traffic police and transport regulators struggle to monitor and control overloading in real time, leading to safety risks and inefficient service management. Additionally, passengers face difficulties in checking ticket prices, as they must manually enter their starting and destination locations to obtain fare details.

1.2 Research Gap

While many digital transport solutions exist around the world, most mainly focus on GPS-based bus tracking and estimated arrival times. However, these systems:

- Rarely provide real-time seat availability or monitor passenger load. This leaves commuters unsure about their comfort and safety.
- Do not have automated fare calculation methods that adjust based on how many passengers board, the trip's length, or demand.
- Fail to effectively combine IoT, AI, and camera monitoring to tackle the problem of bus overloading in real time.
- Offer limited support for decision-making for transport authorities. They lack tools like predictive analytics for optimizing fleets, insights into revenue, and help with policy-making.

1.3 Research Problem

Urban commuters face significant challenges when using public transport due to the lack of real-time information on bus occupancy and ticket fares. Existing systems typically provide only location tracking. They do not indicate whether an approaching bus is overcrowded, partially filled, or has available seating. This uncertainty leads to long wait times, discomfort, and unsafe travel conditions. At the same time, transport authorities and traffic police lack efficient tools to monitor and regulate passenger overloading in real time. This results in safety risks and inefficiencies. Additionally, passengers find it hard to get accurate and instant ticket fare information for their journeys since fare details are not integrated into current systems. There is a clear need for a unified, smart solution that brings together real-time bus tracking, passenger occupancy detection, and automated fare calculation to improve the commuter experience and support effective regulation of public transport operations.

2.0 Objectives

2.1 Main Objective

To make a platform to use public bus service more efficiently. This system provides service to both passenger and driver side.

2.2 Specific Objectives

Performing this research has the following sub-objectives.

- To design and implement a passenger occupancy detection mechanism that provides real-time information on whether a bus is full, partially occupied, or has available seating.
- Create a cloud data pipeline for processing and storing bus data in real time.
- Provide a dashboard on the web or mobile device for administrators and passengers.
- To create a fare calculation feature that allows passengers to easily check bus ticket prices by entering their starting and destination locations.

3.0 METHODOLOGY

3.1 System Overview

The proposed Smart Bus System is a modular and integrated solution designed to enhance public transportation in Sri Lanka through real-time transit optimization and improved passenger experience. The system consists of four key components, each addressing important aspects of urban bus operations by leveraging cutting-edge technologies such as IoT, computer vision, artificial intelligence, and human-computer interaction.

The first component focuses on IoT-based real-time bus tracking and ETA prediction. It utilizes a hybrid connectivity strategy that combines GSM and Wi-Fi to continuously transmit data in diverse environments. Advanced LSTM-based AI models trained on GPS, weather, traffic, and event data enable precise prediction of bus arrival times. Additionally, edge processing on Arduino hardware offers a low-cost, scalable solution suitable for deployment in developing regions.

The second component employs vision-based passenger counting and occupancy monitoring by integrating computer vision and deep learning optimized for embedded systems without GPUs. This module combines binocular depth cameras with thermal sensors to ensure accurate passenger detection while preserving privacy through anonymized data processing.

The third component introduces a mobile-based passenger interaction and accessibility system, emphasizing inclusive design. It supports multi-modal interactions such as audio, vibration, and multilingual features to accommodate differently-abled users. Real-time input through QR codes or a mobile app enables dynamic adjustments to system behavior. Moreover, passengers receive crowd-aware journey recommendations based on live seat availability and occupancy forecasts.

The fourth component is the central fleet management and route optimization engine, which applies data analytics, optimization algorithms, and fleet

intelligence. It dynamically adjusts bus routes and schedules in response to passenger density and traffic, moving beyond static timetables. The AI-powered predictive scheduling with a feedback loop refines operations by learning from historical and current data. This engine also integrates train and ride-sharing services into a unified route planning system, facilitating seamless multimodal travel.

Together, these components create a comprehensive Smart Bus System aimed at improving operational efficiency, reducing passenger waits, enhancing accessibility, and promoting sustainable urban mobility. The modular design allows for parallel development and scalable deployment, addressing the multifaceted challenges of public transit in Sri Lanka through innovative technology and user-centered solutions.

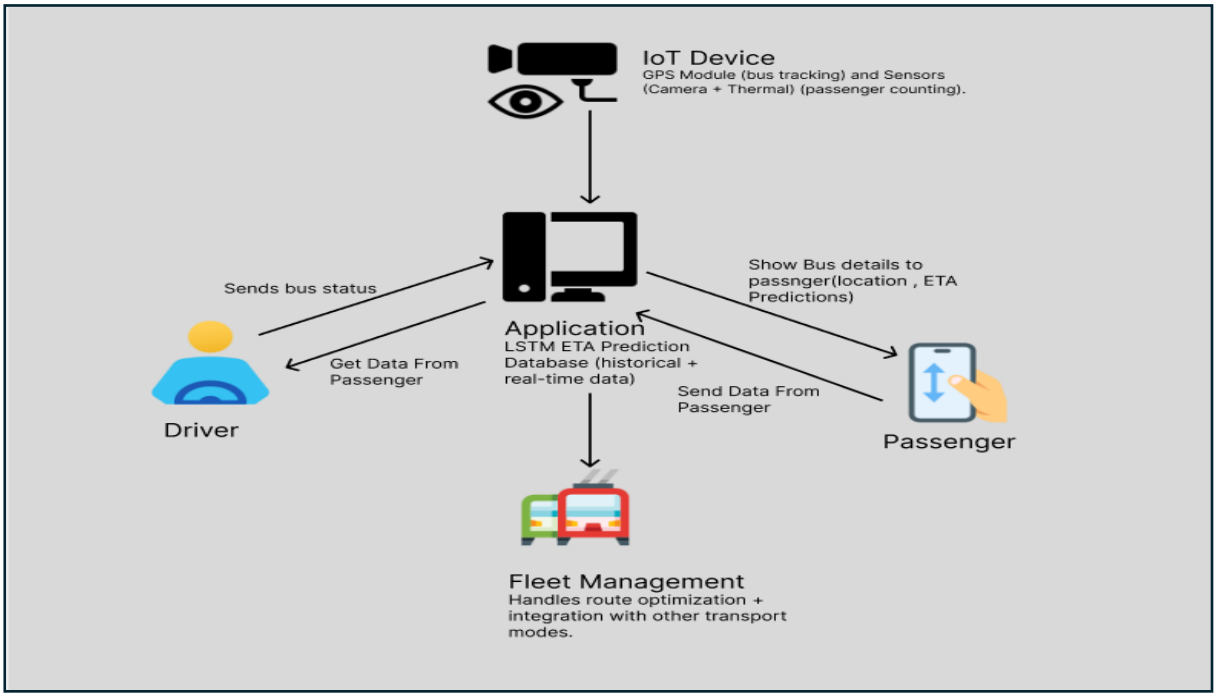

Figure 1: System Diagram

3.2 Individual Component

3.2.1 System Overview

Urban commuters often face significant uncertainty when using public transport. Passengers are unable to know in advance whether an approaching bus is already full, partially occupied, or has available seating. This lack of visibility leads to long waiting times, discomfort, and in many cases, unsafe overcrowding.

On the other hand, authorities such as traffic police and transport regulators struggle to monitor and control overloading in real time, which poses safety risks and reduces the efficiency of transport services.

To address these issues, a smart transport system can be introduced where commuters are able to:

- Check real-time seat availability (full, partially filled, or vacant) before the bus arrives.
- View ticket prices instantly by entering their starting location and destination.
- Identify the best travel options, such as buses with more seat availability, affordable pricing, and shorter waiting times.

This solution not only improves the overall passenger experience by making travel more convenient and safer but also enables transport authorities to manage fleets more effectively, ensuring safer, more reliable, and more efficient public transport services.

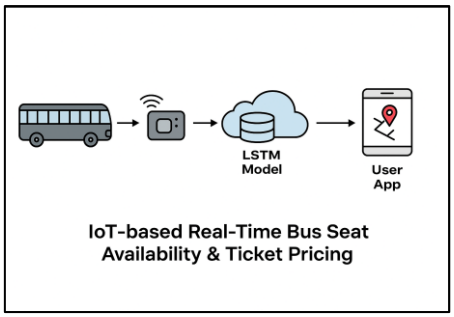

Figure 2: Individual Component System Diagram

3.3 Requirement Gathering

Requirement gathering will involve:

- Reviewing existing literature on computer vision and machine learning[6].
- Studying public transport challenges in the Sri Lankan context. • Interviewing bus operators and passengers.

3.3.1 Past Research Analysis

In terms of previous research analysis, the subjects of bus tracking systems and ETA prediction are covered in a sizable number of research papers and publications. Nonetheless, a lot of articles took the ETA Prediction into consideration.

Finding the methods and resources used to create all of the current tools and platforms was the primary goal of the previous study investigation. Additionally, it aided in identifying the issues that earlier researchers encountered.

3.4 Feasibility Study

3.4.1 Technical Feasibility

Technologies & Tools:

- TensorFlow or PyTorch for training occupancy detection models.
- GSM (SIM800L) and Wi-Fi connectivity
- Cloud database (Firebase, AWS DynamoDB)
- OpenCV for camera-based people counting.
- ReactJS for dashboard UI

3.4.2 Schedule Feasibility

This is an independent solution as a part of Sparkle Edu. This will be developed according to the timeline. Figure 6 and Table 2 illustrates the schedule management plan clearly.

3.4.3 Economic Feasibility

The system will use low-cost IoT hardware and scalable cloud resources with pay-as you-go billing.

3.5 System Analysis

3.5.1 Software Solution Approach

The methods for developing the answer are listed below, and figure 5 shows the whole system flow:

1. (Collect **passenger occupancy data** using smart sensors or cameras).
2. Data Acquisition: Use Wi-Fi or GSM connectivity to send the collected data from IoT devices to a cloud-based database.
3. Preprocessing: Apply preprocessing techniques to the gathered data, including time-series alignment, handling missing values, clearing inaccurate counts and seat availability, and normalizing input features (seat availability, crowd calculation, etc.).
4. Feature Extraction: From the preprocessed dataset, extract relevant features for ETA prediction. Some examples of features are:
    a. Real time passenger count.
    b. Seats availability.
    c. Generate ticket prices.
    d. Offer best routes which have max seat availability.

5. Deploy the trained LSTM model in the cloud, integrate it with the IoT data pipeline, and provide real-time ETA predictions to passengers via a web/mobile application.

3.5.2 Tools & Technology

- OpenCV for camera-based people counting.
- TensorFlow or PyTorch for training occupancy detection models.
- GSM (SIM800L) and Wi-Fi connectivity .
- Cloud database (Firebase, AWS DynamoDB).
- Python-based LSTM model (TensorFlow/Keras).
- ReactJS for dashboard UI.
- GitHub for version controlling.

3.6 Project Requirements

3.6.1 Functional Requirements

- Real-time crowd calculation and seat availability.
- Generate ticket price according to the distance.
- Offer best routes which have max seat availability.

3.6.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested model are listed below.

- Usability

- Reliability
- Availability
- Accuracy
- Performance

3.7 Testing

It is essential to the effectiveness of this research when it comes to testing the findings. The first plan is to test the solution among a few cars in order to determine whether the suggested remedy works. We can determine if the device is functioning or not.

Nevertheless, this technique will be tested using the chosen bus services in the last phase of the study. This will function similarly to acceptance testing (beta, alpha). The majority of the testing will be done by hand.

Test Cases

Test Case ID	**Description**	**Prediction**	**Test Steps**	**Expected Result**
TC-1	Check real-time bus seat availability	System should show seat count (Full / Partially Filled / Available)	1. Open the app/website 2. Search for a bus route 3.Select an upcoming bus	Seat availability status is displayed correctly in real time

TC-2	Check ticket price calculation	Correct ticket price should be displayed based on start and end location	1. Enter”Colombo” as start location 2. Enter “Kandy” as destination 3.Click "Check Fare"	Ticket price is displayed correctly.
TC-3	Find best route with maximum available seats	System should recommend a bus with more available seats	1. Enter journey details (start & end location) 2. Click "Find Best Route" 3. View results	Bus with maximum available seats is highlighted /recommended
TC-4	Check overloading alert for authorities	System should notify when bus occupancy > 100%	1. Track a bus with sensor data showing passenger count exceeding seat capacity	Authority panel shows “Overloaded” alert

			2. Refresh bus status	

TC-5	Verify seat update when passengers board/exit	System should update seat count in real time	1. Track a bus with 20/50 seats occupied 2. Add 10 passengers (board) and remove 5 passengers (exit) 3. Refresh availability	New seat count = 25/50 occupied

Table 1: Test Cases

3.8 Timeline

The proposed timeline for the project is as follows.

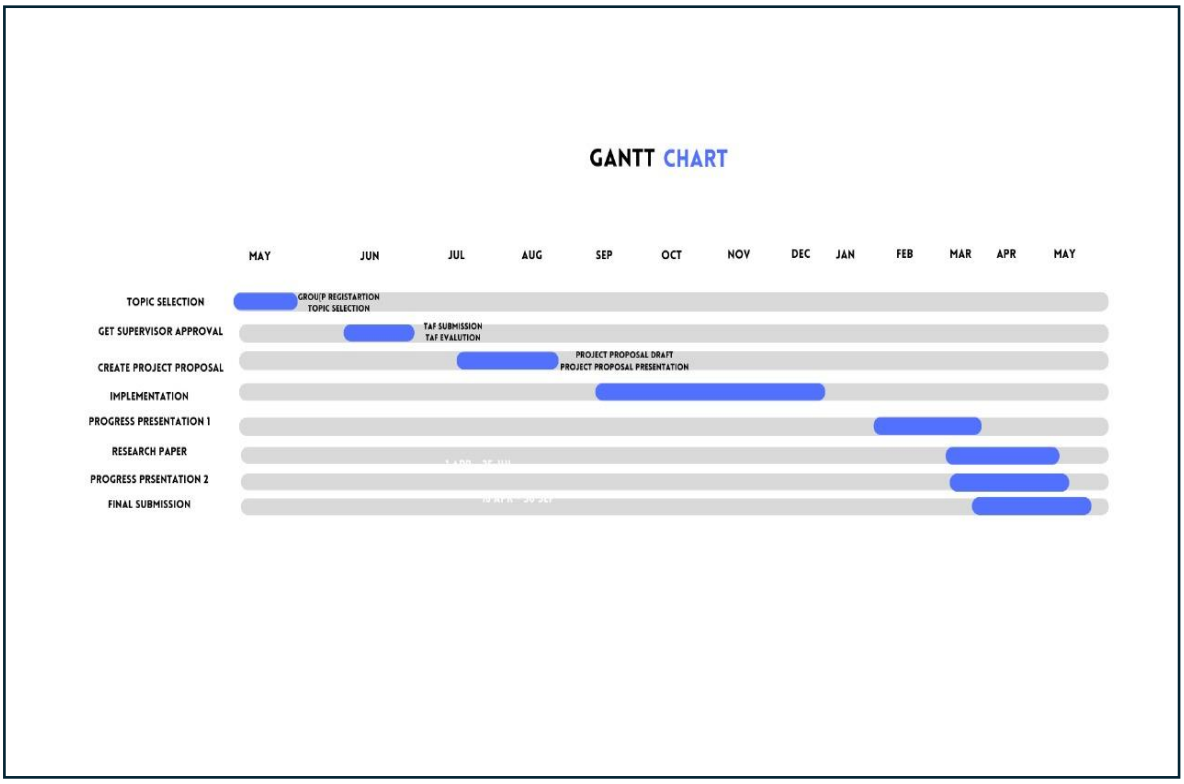

Figure 3: Gantt Chart for Project Schedule Management

3.9 Communication Management Plan

The Communications Management Plan ensures that the team, supervisor, and co-supervisor, among other project stakeholders, receive the information they require to perform their duties effectively throughout the project. For the project to be successful, communication activities need to be well planned and executed.

The most efficient and successful methods of communicating with different stakeholders are described in this strategy. The target audience, content, format, frequency, and expected outcomes of each communication channel are all described and documented. It also describes each stakeholder's responsibilities, task-assignment procedures, and communication strategies based on their level of interest, influence, and project expectations.

The key objectives of this communication plan are:

- To ensure timely and accurate flow of information among all stakeholders.
- To promote transparency and accountability within the project team.
- To facilitate feedback loops that help in improving project deliverables.
- To maintain alignment between project goals, supervisor guidance, and team execution.

3.9.1 Communication Objectives:

Proactive communication is important on all projects. Communication needs to be:

- Adequate: in the right format and right content.
- Specific: for the targeted audience.
- Sufficient: providing all the necessary information.

- Concise: brief, avoiding repetition and non-important information.
- Timely: addressing points at the right time.

3.9.2 Communication Media:

The communication media that will be used for the project are:

1. **Email** – For official communications, documentation, and approvals.
2. **Documents (MS Word / PowerPoint)** – For sharing project reports, proposals, and presentations.
3. **Instant Messaging (WhatsApp, MS Teams Chat)** – For day-to-day updates and quick clarifications.
4. **Meetings (Physical or MS Teams Virtual Meetings)** – For project discussions, milestone reviews, and progress presentations.
5. **Phone Calls** – For urgent communications requiring immediate response.

4.0 COMMERCIALIZATION

The Smart Bus Transport System has significant potential for commercialization. It addresses important issues for commuters and authorities by providing real-time seat availability, fare calculation, and monitoring of overloading. The system can generate revenue through several models, including subscription plans for passengers, commissions from digital ticket sales, partnerships with bus operators for fleet management dashboards, targeted in-app advertising, and data analytics services for authorities. A step-by-step implementation strategy can be used, starting with a pilot deployment on selected urban routes, then expanding to intercity and regional services, and eventually achieving nationwide coverage with partnerships from the government and operators. Its competitive advantages include offering real-time transparency, improving passenger safety, and enabling efficient transport management. With the opportunity to be a first mover in Sri Lanka, the system can also grow into regional markets, making it both socially impactful and commercially viable.

5.0 BUDGET

Since the outcome of the proposed model is a software-based solution, there are no hardware components connected to the implementation. The primary source of the cost will be the subscription fees to the cloud provider for the computing power of the virtual machines. However, there will be some other costs expected to as be given in the table below.

Type	Cost
IoT hardware (Arduino/ESP32, GPS, GSM module, sensors) per bus	10000 LKR
Cloud hosting & database services (annual)	12100 LKR
Internet/data SIM charges(Aproximately) (per bus annually)	6000 LKR
Stationery, documentation & miscellaneous	3000LKR
TOTAL	***31100 LKR***

Table 2: Cost Management Plan

This amount may differ according many reasons.

6.0 SUMMARY

The Smart Bus Transport System can be marketed through subscriptions, digital ticket fees, partnerships with operators, and advertising. It offers real-time seat availability, clear fare information, and safety alerts. This system benefits both commuters and authorities. It is a scalable and practical solution for Sri Lanka and other regions..

SMART BUS SYSTEM FOR REAL-TIME PASENGER MANAGEMENT SYSTEM

25-26J-267

Project Proposal Report

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Supervised by: Nelum Amarasena

B.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor:

(Mr. Nelum Amarasena) Date: 29/08/2025

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Signature of the co-supervisor:

(Mrs. Uthpala Samarakoon) Date: 29/08/2025

Abstract

This component focuses on developing a mobile-based passenger interaction system that redefines accessibility and engagement within smart public transit. The system provides a multi-modal interface that is adapted to the demands of a variety of users, including those with impairments, by utilizing inclusive design and human-computer interaction concepts. This includes multilingual choices, vibrations, and auditory cues. In addition to providing basic information, the platform enables real-time feedback and demand-responsive routing changes through mobile engagements including QR code scanning and app inputs, enabling passengers to actively engage in transit planning. Passengers also receive dynamic route recommendations that consider seat availability and crowd levels, improving comfort and the entire travel experience. This integrated strategy helps create a smart bus system that is more effective, flexible, and user-cantered while also enhancing accessibility and usability.

Keywords: Mobile-Based Passenger Interaction, Smart Public Transit, Human-Computer Interaction, Multi-Modal Interface, Multilingual Support, Real-Time Passenger Feedback

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ETA	Estimated Time of Arrival
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
IoT	Internet of Things
LSTM	Long Short-Term Memory (neural
MQTT	Message Queuing Telemetry Transport
QR	Quick Response
RESTful API	Representational State Transfer API
UI	User Interface
UX	User Experience
VR	Vibration Response
WIFI	Wireless Fidelity

Table 1: List of Abbreviations

1.0 INTRODUCTION

1.1 Background and Literature

With growing urbanization and the increasing demand for environmentally friendly mobility, modern public transportation systems must be more flexible, inclusive, and user-friendly than traditional models. The way passengers interact with these systems and access real-time information is just as critical as route optimization and vehicle tracking. Accessibility and mobile-based passenger engagement play a vital role in ensuring that smart bus systems cater to diverse passenger needs [1][2].

A well-designed mobile application can bridge the gap between passengers and complex transit operations. Inclusive design and human-computer interaction (HCI) principles ensure usability across different skill levels and language preferences, while accessibility features such as vibration alerts, audio guidance, and multi-language support improve the travel experience for passengers with disabilities [1][2]. Recent studies highlight that enhancing accessibility is essential for promoting equity in transport services [2].

Mobile systems also enable interactive demand-responsive routing, empowering passengers to influence travel plans dynamically. This flexibility is especially beneficial during peak hours or unexpected service disruptions. Prior research shows that mobile-enabled interaction significantly enhances passenger satisfaction and decision-making by delivering “mobile moments” that personalize the journey [3]. Complementary features such as crowd-aware trip suggestions provide real-time insights into seat availability and bus capacity, supporting informed travel choices [4][5].

Beyond accessibility and engagement, emerging technologies such as computer vision and stress-detection mechanisms can further enhance passenger experience. Vision-based systems can monitor facial expressions and body signals to detect stress, fatigue, or discomfort, thereby contributing to safer and more adaptive transportation services [6][7][8]. Mobile-based stress monitoring also offers non-invasive ways to assess passenger well-being during trips [9]. Furthermore, advanced classification techniques that combine multiple features (e.g., facial expressions, body parameters, and contextual cues) have been shown to improve recognition accuracy in human-computer interaction systems [10].

Altogether, this component not only improves travel experiences but also supports the broader objective of developing an intelligent, adaptive, and socially inclusive public transport ecosystem that integrates accessibility, real-time interaction, and future-ready AI enhancements.

Figure 1: Key Accessibility Features in Mobile UI Design

1.2 Research Gap

The effectiveness and inclusiveness of smart bus systems are limited by a few unsolved issues, despite notable technical developments. Existing IoT-based tracking and ETA prediction systems sometimes lack thorough integration with other components for seamless operation, making it difficult to sustain continuous, context-aware data transmission in a variety of urban situations. The practical implementation of vision-based passenger counting algorithms on embedded systems is limited due to their large processing requirements and frequent privacy sacrifices, despite their rising accuracy.

Furthermore, accessibility features and passenger engagement on mobile devices typically lack strong multi-modal interfaces that can accommodate individuals with disabilities inclusively through multilingual, haptic, and auditory support. In addition, these systems seldom include real-time passenger input for demand-responsive routing or offer dynamic itinerary recommendations based on crowd density and seat availability. Most current fleet management solutions lack real-time, demand-aware route planning that integrates multi-modal transportation modes for comprehensive urban mobility and instead rely on static timetables.

Additionally, implementing fully integrated smart bus systems is made more difficult by growing worries about cybersecurity flaws, data privacy, and the lack of clear regulatory requirements. Therefore, it is imperative to close these gaps using modular, interoperable solutions that improve security, privacy, inclusiveness, and real-time responsiveness in smart public transit environments.

Product/Platform	Multi-Modal Accessibility (Audio, Vibration, Multilingual)	Real-Time Passenger Interaction (Tracking, Booking, QR)	Demand Responsive Routing	Crowd-Aware Journey Suggestions	Inclusivity for Differently Abled
Google Maps Transit	No	Limited (Tracking only)	No	Limited	No
Transit App	Limited (Tracking, schedule)	Limited (Tracking, schedule)	No	Limited	No
Local Bus App (Sri Lanka)	No	Limited (Tracking only)	No	No	No
Proposed Mobile Passenger System	Yes	Yes	Yes	Yes	Yes

Figure 2: Competitive Analysis

1.3 Research Problem

Despite significant advances in smart bus systems, key challenges remain in ensuring accessibility, real-time passenger engagement, and effective use of dynamic data. Current systems often fail to meet the unique needs of passengers with disabilities and lack inclusive mobile interfaces that support universal accessibility [1][2].

Moreover, existing platforms rarely provide interactive, demand-responsive routing and crowd-aware trip suggestions. Instead, they rely heavily on static schedules, overlooking the benefits of dynamic passenger participation and mobile-based travel assistance [3][4][5]. This restricts passenger autonomy and overall satisfaction.

Another limitation is the lack of comprehensive, real-time insights into bus occupancy levels. Current systems seldom allow passengers to indicate onboard congestion, estimate seat availability, or select a bus based on live conditions [4]. This gap results in uninformed decision-making, reducing passenger comfort and trust in public transport.

Beyond operational issues, smart transit systems have not fully embraced vision-based and AI-driven approaches that could improve safety and passenger well-being. For instance, stress detection using facial expressions, body signals, and mobile-based sensors could provide valuable insights into passenger states during travel [6][7][8][9][10]. However, these techniques are rarely integrated into transport frameworks, limiting the potential of intelligent, passenger centered services.

Additionally, data privacy concerns cybersecurity risks, and the absence of standardized protocols for safe, interoperable communication between transport components hinder the creation of robust urban mobility ecosystems.

To address these gaps, this project proposes a modular smart bus system that combines inclusive mobile applications, IoT-enabled real-time tracking, vision-based occupancy monitoring, and AI-driven passenger well-being assessment. By enabling passengers to engage actively in trip planning, view bus arrival and occupancy details, and even report congestion in real time, the system empowers users to make informed travel choices. Combined with dynamic fleet management that adapts routes based on real-time traffic and passenger inputs, the proposed approach aims to deliver a sustainable, efficient, and socially inclusive urban transport environment.

2.0 Objectives

2.1 Main Objective

To design and develop a complete Smart Bus System that improves urban public transportation by providing real-time bus tracking and accurate arrival information, monitoring bus occupancy while respecting passenger privacy, enabling better communication and interaction with passengers including those with disabilities and managing the entire fleet more effectively through flexible route and schedule adjustments based on actual passenger demand and traffic conditions. This system aims to make bus travel more efficient, comfortable, and accessible for everyone.

2.2 Specific Objectives

The following are the sub-objectives of conducting this research.

- To develop a user-friendly mobile application that provides passengers with access to real-time information about bus arrivals and occupancy, assisting them in making decisions about whether to board right away or wait for the next bus.
- To enable passengers to actively share information about the number of people on the bus through the app, improving the accuracy of crowding information for all users.
- To allow users to submit requests and feedback in real time via the app or QR codes, enabling the transportation system to dynamically modify routes and services in response to demand.
- To use intelligent data analysis to interpret passenger inputs and occupancy trends to continuously improve route planning and passenger experience while safeguarding privacy.

3.0 METHODOLOGY

3.1 System Overview

The proposed Smart Bus System is a modular and integrated solution designed to enhance public transportation in Sri Lanka through real-time transit optimization and improved passenger experience. The system consists of four key components, each addressing important aspects of urban bus operations by leveraging cutting-edge technologies such as IoT, computer vision, artificial intelligence, and human-computer interaction.

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The second component employs vision-based passenger counting and occupancy monitoring by integrating computer vision and deep learning optimized for embedded systems without GPUs. This module combines binocular depth cameras with thermal sensors to ensure accurate passenger detection while preserving privacy through anonymized data processing.

The third component introduces a mobile-based passenger interaction and accessibility system, emphasizing inclusive design. It supports multi-modal interactions such as audio, vibration, and multilingual features to accommodate differently abled users. Real-time input through QR codes or a mobile app enables dynamic adjustments to system behavior. Moreover, passengers receive

crowd-aware journey recommendations based on live seat availability and occupancy forecasts.

The fourth component is the central fleet management and route optimization engine, which applies data analytics, optimization algorithms, and fleet intelligence. It dynamically adjusts bus routes and schedules in response to passenger density and traffic, moving beyond static timetables. The AI-powered predictive scheduling with a feedback loop refines operations by learning from historical and current data. This engine also integrates train and ride-sharing services into a unified route planning system, facilitating seamless multimodal travel.

Together, these components create a comprehensive Smart Bus System aimed at improving operational efficiency, reducing passenger waits, enhancing accessibility, and promoting sustainable urban mobility. The modular design allows for parallel development and scalable deployment, addressing the multifaceted challenges of public transit in Sri Lanka through innovative technology and user-cantered solutions.

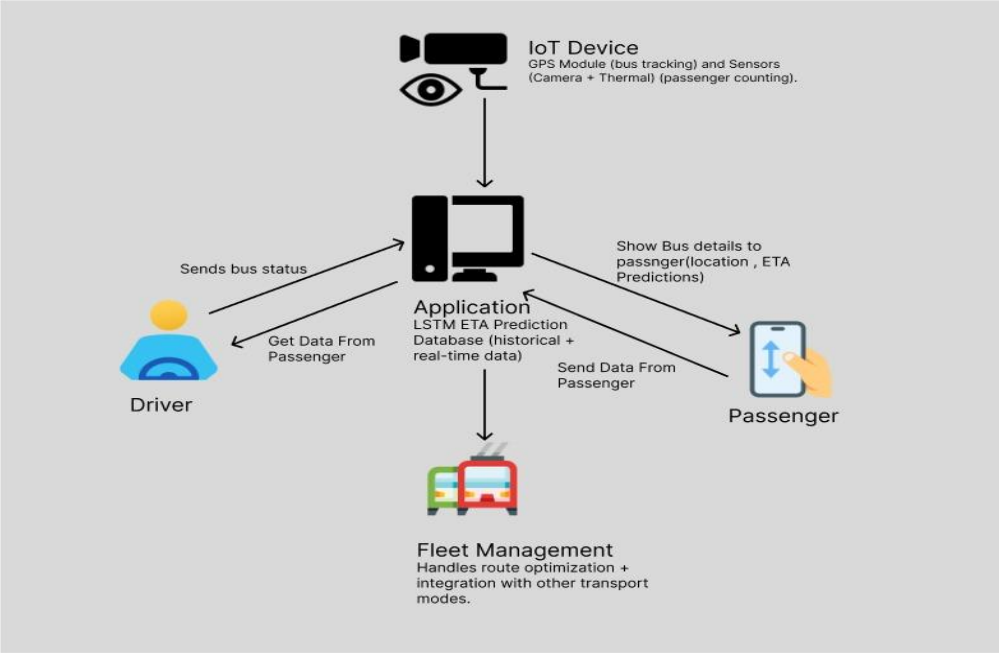

Figure 3: System Diagram

3.2 Individual Component Overview

A platform that is inclusive and engaging, the Mobile-Based Passenger Interaction and Accessibility System was created to improve the Smart Bus System's passenger experience and participation. To especially serve users with disabilities and a variety of passenger demographics, this component promotes accessibility by providing multi-modal interaction techniques such as audio assistance, vibrating notifications, and bilingual interfaces.

Passengers may provide real-time information on bus occupancy, preferred routes, and comments on transit services using a smartphone app or QR code scanning capability. By adjusting route recommendations and itinerary choices based on real-time seat availability and anticipated crowding, the system dynamically responds to this information. By assisting passengers in making well-informed decisions, such as whether to take the present bus or wait for a less packed one, this crowded-aware route eventually improves comfort and autonomy.

Data communication with backend services is secured and optimized for real-time responsiveness, ensuring that passenger inputs are promptly processed. The system protects privacy by anonymizing submitted user data, while intelligent analysis of passenger feedback contributes to improving fleet management and route optimization in coordination with other system modules.

This component acts as the critical touchpoint between passengers and transit operations, blending human-computer interaction principles with adaptive technologies to deliver a seamless, accessible, and responsive public transportation experience tailored to the needs of all users.

3.3 Feasibility Study

3.3.1 Technical Feasibility

3.3.1.1 Knowledge of Technologies

To develop the proposed solution, the knowledge in below mentioned technologies should be wanted.

1. Mobile App Development: Expertise in building responsive, multi-platform mobile applications using Android/iOS native technologies or cross-platform frameworks (such as Flutter or React Native).
2. Human-Computer Interaction (HCI) and Accessibility: Understanding inclusive design principles to support multi-modal interactions like audio, vibration, and multilingual interfaces, ensuring usability for differently abled passengers.
3. Real-Time Communication Protocols: Knowledge of WebSocket, MQTT, and HTTP/HTTPS protocols to enable low-latency data exchange between the app and backend.
4. Backend Development: Familiarity with RESTful API or GraphQL design, backend frameworks (Node.js, Django, Flask), and secure data handling including encryption and anonymization.
5. Cloud and Database Services: Experience with cloud platforms such as Firebase, AWS, or Google Cloud for hosting APIs and real-time databases.
6. Data Analytics Fundamentals: Basic knowledge of how passenger feedback and occupancy data can be processed to inform dynamic route adjustments and journey recommendations.
7. Testing and Deployment: Skills in mobile app testing (unit, UI, accessibility testing) and knowledge of deployment processes for Google Play Store and Apple App Store.

3.3.1.2 Data Collection Knowledge

Data collection and preprocessing are essential steps in any data analysis task.

1. Data Collection: IoT Devices and Sensors, Real-Time Monitoring, Communication Protocols
2. Data-Preprocessing: It includes the procedures of data cleaning, data transformation, data reduction, and data integration.

3.4 Tools & Technology

- Programming Languages – JavaScript (React Native)
- Frameworks – React Native
- Database Systems – MongoDB
- Accessibility Libraries – React Native Accessibility API
- UI Design Implementation – Figma
- Diagramming Tool – Draw.io
- Backend & Cloud Services – Firebase or AWS
- IoT Communication – MQTT protocol or HTTP REST APIs
- Integrated Development Environment (IDE) – VS Code
- Version Control System (VCS) – Git (GitHub or GitLab)
- Collaboration Tools – Microsoft Teams or WhatsApp.
- Testing & Accessibility Validation
    - o Automated Testing- Jest (React Native)
    - o Accessibility Testing– Axe or Google Accessibility Scanner
- Deployment Tools
    - o Docker - Containerization
    - o Kubernetes - Orchestration

    - Jenkins - Automation
    - Ansible - Configuration
    - Chef - Infrastructure
- Project Management – MS Planner

3.5 Project Requirements

3.5.1 Functional Requirements

The functional requirements for the proposed model are as follows:

- Real-time bus arrival and occupancy display

  This feature allows passengers to view accurate and up-to-date information about the expected arrival time of buses at their stops and the current occupancy or crowding inside the bus. Utilizing live GPS data and sensor inputs, the system predicts bus locations and arrival times, displaying this on the mobile app interface. It helps passengers make informed decisions, reducing waiting time and avoiding overcrowded buses. The display needs to be visually clear, easily accessible, and frequently updated with minimal latency.

- Real-time passenger feedback submission

  Through the smartphone app or by scanning QR codes on the buses, passengers can immediately give real-time comments on bus occupancy, comfort, and service quality. With the help of this feedback loop, the system may collect crowding data, which increases the accuracy of occupancy data and dynamically notifies fleet management about capacity and route changes. For passengers to actively participate, the submission procedure should be simple, fast, and compatible with a variety of input methods.

- Multi-modal interaction capabilities (audio, vibration, multilingual)

  Multiple interaction modalities that are suited to a range of user demands are supported by the system to guarantee accessibility and inclusivity. Spoken announcements and confirmations based on audio are examples of audio features. Passengers can be quietly alerted via vibration signals, which are helpful for visually challenged people. Multilingual support accommodates non-native speakers by offering the app's interface and interactions in many languages. The improved user experience and wider accessibility offered by these multi-modal choices are guaranteed for all passenger groups, including those with disabilities.

3.5.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested model are listed below.

- Usability
  - The user interface of the mobile app must be simple and straightforward, making it suitable for users of all ages and skill levels, including travelers with disabilities.
  - Accessibility features such as audio guidance, vibration alerts, multilingual support, and high-contrast modes should be implemented.
- Reliability
  - During regular use, the system must function reliably and manage passenger interactions without any issues.
  - Real-time data updates and feedback submissions shall be accurately processed and reflected in the system without loss.

- Availability
    - The app and backend services will be accessible 24/7 with minimal downtime, ensuring passengers can rely on the system anytime.
- Accuracy
    - Passengers must be presented with accurate information on bus arrival timings, capacity levels, and suggested routes based on the most recent data gathered from sensors and passenger inputs.
- Performance
    - The system must react to human interaction and provide real-time data in a reasonable amount of time.
    - The application must continue to provide a responsive and seamless user experience even when there is a significant user load.

3.5.3 User Requirements

With an emphasis on essential functionality and user demands, here are the user requirements:

- Real-Time Information Access
    - The system shall provide passengers with real-time updates on bus arrival times and current occupancy levels
    - The information must be displayed clearly on the mobile app interface.
- Multi-Modal Interaction Support
    - The app should support multiple modes of interaction, including audio prompts, vibration alerts, and visual displays, to accommodate differently abled users.
- Passenger Feedback Submission
    - Passengers will be able to use the app or QR code scanning to provide real-time feedback on bus congestion and service quality.

    - Feedback submission should be simple and quick to encourage frequent use.
- Privacy and Security
    - To safeguard user privacy, the system will anonymize all data given by passengers.
    - Encryption techniques will be used to protect data transit between the application and backend servers.
- Ease of Use and Accessibility
    - The app interface should be intuitive and easy to navigate for all groups and abilities.
    - It must include accessibility features like screen reader compatibility and high contrast options.

3.6 Testing

Test Case ID	Description	Preconditions	Test Steps	Expected Results
T01	Verify mobile app displays real-time bus arrival info	User is logged into the app	1. Open the app 2. Select a bus route 3. Observe the displayed bus arrival time	Real-time bus arrival time is accurately displayed without delay
T02	Verify passengers can report to the bus station in the app	App is open and bus route selected	1. Navigate to occupancy feedback section 2. Submit the number of passengers currently on board	Bus station data is successfully submitted and reflected in app
T03	Verify multi-modal interaction support	User has accessibility settings enabled (audio/vibration)	1. Enable audio/vibration options 2. Navigate through app features using these modes	App responds correctly to audio and vibration interactions
T04	Verify real-time route updates based on passenger feedback	App is running and user feedback submitted	1. Submit a route change request via app 2. Check if the route updates timely	Successfully added routes adjust
T05	Verify crowd in the bus	App running with real-time occupancy data	1. Select a journey plan 2.Select load and seat availability	Can review current crowding and seat availability

T06	Verify vibration feedback on bus arrival notification	Vibration enabled on device	1. Enable vibration 2. Wait for bus arrival alert via app	Device provides vibration alert when bus arrival notification is received
T07	Verify audio assistance works for differently abled users	Audio assistance enabled	1. Enable audio assistance 2. Navigate app menus and notifications	Audio prompts play correctly and guide the user
T08	Verify privacy of passenger feedback submission	User logged in	1. Submit occupancy or feedback data	User data is anonymized and securely transmitted
T09	Verify user can customize notification preferences	User logged in	1. Go to notification settings 2. Enable/disable various notifications	User preferences are saved, and get notifications accordingly
T010	Verify app works offline with latest cached data	App previously synced with data	1. Turn off internet 2. Open app and view bus arrival and occupancy info	App displays last available data without errors

Table 1: Expected Test Cases Table

3.7 Timeline

The proposed timeline for the project is as follows.

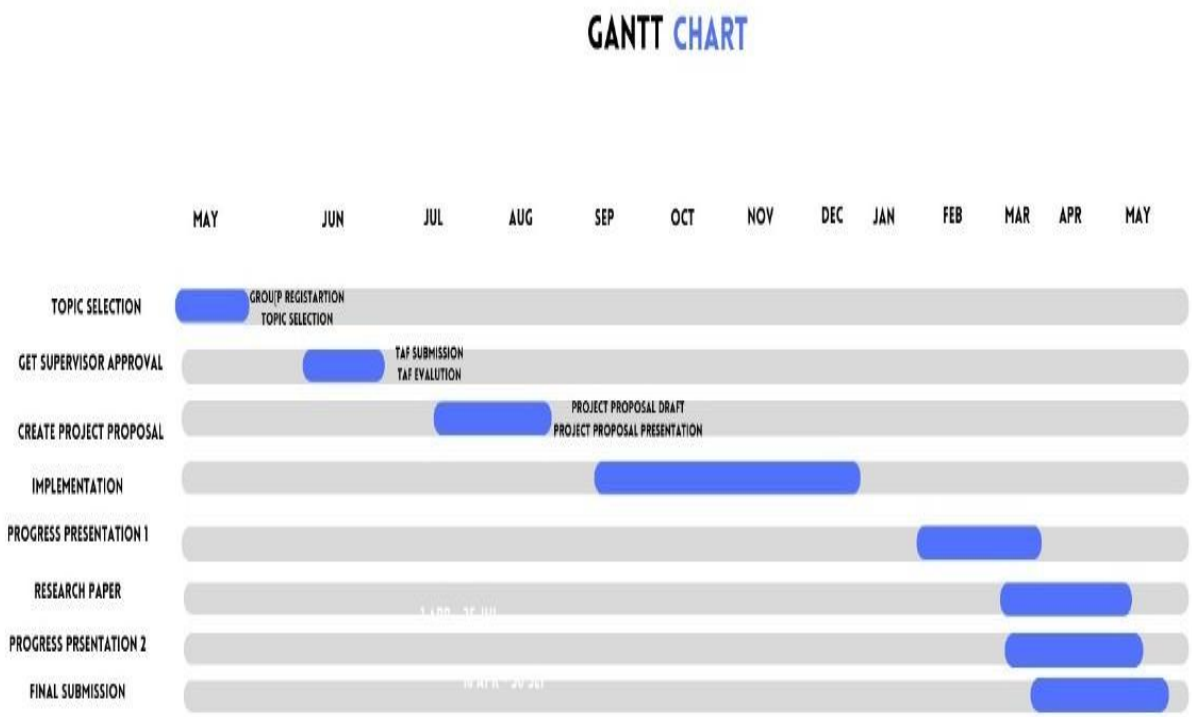

Figure 4: Gantt Chart for Project Schedule Management

3.8 Communication Management Plan

The Communications Management Plan helps to ensure that all team members, supervisor and co-supervisor have the information they need to perform their roles throughout the project. Project success depends on the planning and execution of communication efforts.

How to communicate with different stakeholders most effectively and efficiently is decided by the communications management plan. It outlines and documents the audience, communication items' content, format, frequency, and anticipated outcomes. Also, it specifies the various stakeholders' roles in the project, how to assign tasks to them, and the communication strategy for each of them based on their influence on the project, interests, and expectations.

3.8.1 Communication Media

The communication media that will be used for the project are:

1. Email
2. Document (MS Word and/ PowerPoint)
3. Phone call
4. Meetings (using meeting rooms, conference phones, Ms Teams)
5. Chats (WhatsApp)

4.0 WORK BREAKDOWN CHART

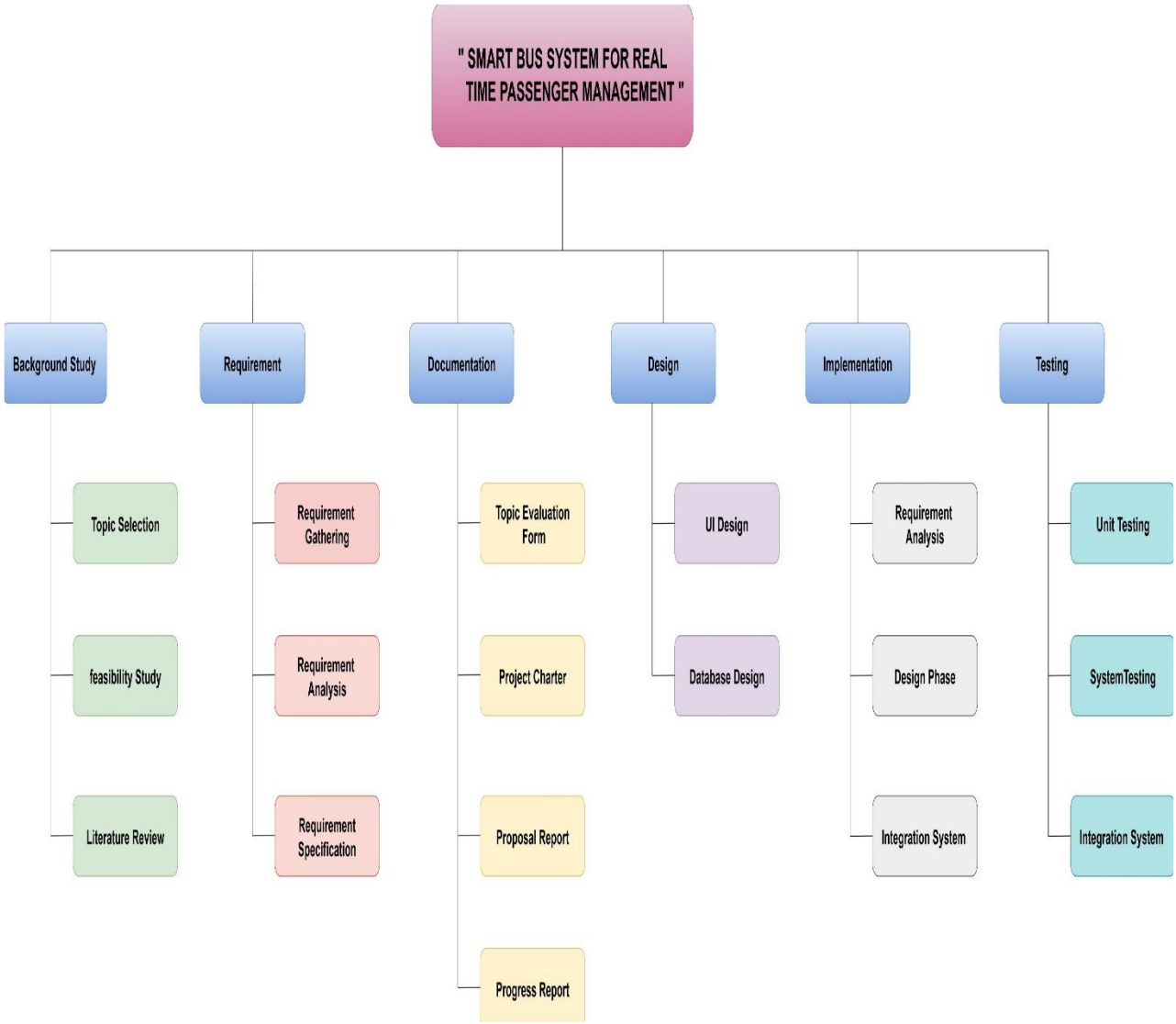

Figure 5: Work Breakdown Chart

5.0 COMMERCIALIZATION

The Mobile-Based Passenger Interaction and Accessibility System stands out as a game changer in how passengers engage with their transportation alternatives as cities expand, and the significance of accessible, environmentally friendly public transit grows. With the help of intelligent technology that listens to and changes with its users, this system offers a customized experience that goes beyond simple tracking and scheduling.

Designed with empathy and inclusion in mind, the app's AI-enabled features—like spoken instructions, gentle vibration alerts, and support for multiple languages—make public transit easier and more welcoming for everyone, especially people with disabilities or those who speak different languages. This inclusive approach isn't just good for passengers; it opens doors for transit agencies to reach wider audiences and build deeper trust within their communities.

Real-time feedback and dynamic routing mean the system reacts swiftly to changing conditions whether it's rush hour traffic or unexpected delays ensuring journeys are smoother and more comfortable. This responsiveness helps reduce overcrowding and waiting times, ultimately improving passenger satisfaction and boosting ridership.

From a commercial standpoint, this solution gives transport operators a means to differentiate themselves by demonstrating their concern for the experiences of all passengers while simultaneously streamlining operations with astute data insights. This technology can grow through collaborations with local governments and tech companies, starting with trial projects in strategic cities. This will encourage innovation and propel long-term advancements.

In essence, this system blends cutting-edge AI with a human touch, shaping a future where public transportation is not only efficient but truly accessible and user-friendly for all.

6.0 BUDGET

Since The budget required for developing the Mobile-Based Passenger Interaction and Accessibility System does not significantly differ from that of typical mobile application projects. Like most mobile apps, costs primarily depend on the complexity of features, platform support (Android, iOS, or both), design requirements, and backend infrastructure. Though tailored for accessibility and real-time interaction, the development approach follows standard practices involving UI/UX design, coding, testing, and deployment phases. Consequently, standard mobile app cost estimates and timelines serve as a good benchmark for planning and budgeting this component.

Type	Cost
Internet use and app hosting	5000 LKR
Mobile App Development Tools & Licenses	12000 LKR
Cloud Subscription	15000 LKR
Publication & Documentation costs	10000 LKR
stationary and Miscellaneous Expenses	2000 LKR
TOTAL	***44000 LKR***

Table 2: Cost Management Plan

7.0 SUMMARY

This part of the Smart Bus System is focused on making passengers' lives easier and more comfortable through a simple, interactive mobile app. It helps users get up-to-date information about bus arrivals and how crowded the buses are, so they can decide the best time to travel. The app is designed to be accessible to everyone, including people with disabilities, by offering audio guidance, vibration notifications, and support for multiple languages.

Passengers can share their feedback and real-time information about bus occupancy using the app or by scanning QR codes, which helps the system adjust routes and schedules on the fly. Behind the scenes, intelligent data processing analyses this input to improve travel suggestions and overall service without compromising passengers' privacy. This makes travel more predictable, inclusive, and stress-free for everyone who uses the buses.

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